

Practical 01: EEG data analysis

```
In [1]: # Credits: this notebook was written by Ilaria Gavetti for the Cognitive
         Neuroscience EEG practical & EEGLab.

         # MNE-toolbox was used as reference to develop and implement the code.
```

PLASE ADD HERE BELOW THE NAMES AND U-NUMBERS OF ALL THE PEOPLE IN YOUR GROUP WHO PARTICIPATED IN THE PRACTICAL

```
In [2]: # NAMES AND U-NUMBERS HERE:
        """
        Francesco Fumagalli 704513
        Cesare Maria Dalbagno 933585
        Mihnea Angeluta 570905
        Chris Martin 653436
        Xavier Krsinar 688637
        """
```

```
In [3]: # ! pip install mne
```

```
In [4]: import mne
         import os
         import numpy as np
         import csv
         import pandas as pd
```

Welcome to the data-analysis part of the EEG pratical. At this point, you should have:

1. collected your data
2. received an output csv file from the data collection procedure, one for each condition, for instance: Open_eyes_1.csv, Open_eyes_2.csv, Closed_eyes_1.csv, Closed_eyes_2.csv.

Follow the instructions below to properly upload and conduct your data-analysis:

Step 0: upload the files

First, I suggest creating a "base" directory, called "EEG practical".

Then, please upload all the .csv files from your laptop in the directory `"../EEG_practical/1. Raw files"`, where the `"..."` part is your personal path to get to **"EEG_practical"** and **1. Raw files** is a folder in this directory which you should manually add.

Also, since we will be saving intermediate files during the analysis, add the following folders to your **"EEG_practical"** directory:

2. Loading & Digitization

3. ICA

4. Epoch (with inside other 3 folders called **Annotations**, **Closed** and **Open**)

5. TFR

The code you will run already has these directories implemented.

Last, but not least, change your current working directory to **"EEG_practical"** if it's not already set to it, to ensure the code runs smoothly.

In [5]:

```
os.getcwd()
```

Out[5]:

```
'C:\\Users\\IamAh\\Desktop\\UNIVERSITY - CSAI\\Cognitive Neuroscience\\M001_Cognitive_Neuroscience_Practical_Material\\EEG_practical'
```

In [6]:

```
# os.chdir("../EEG practical") # add your own path in the "..." part  
if your current working directory is not already the EEG practical  
folder.
```

Steps 1&2: Loading: you might have to troubleshoot a bit with the code to adjust the directories from your enviroment, so don't worry if you get some errors about that at the beginning.

For .csv extension

In [7]:

```
# Read the CSV file as a NumPy array

for file in os.listdir("1. Raw files"):
    if "csv" in file:
        data = pd.read_csv("1. Raw files/"+file, delimiter=",", verbose
= True)
        data_sub = data[["EEG 1", "EEG 2", "EEG 3", "EEG 4", "EEG 5",
"EEG 6", "EEG 7", "EEG 8"]] # only select the electrodes columns, not the
rest

        # Print file name
        print("*****"+file+"*****")

        # Some information about the channels
        ch_names_new = ["Fz", "C3", "Cz", "C4", "Pz", "P07", "Oz",
"P08"] # from Unicorn BCI manual

        # Sampling rate of the Unicorn BCI
        sfreq = 250 # Hz

        # Create the info structure needed by MNE
        info = mne.create_info(ch_names_new, sfreq, ch_types =
["eeg"]*len(ch_names_new), verbose = True)

        # Set montage
        info.set_montage(montage = 'standard_1020')

        # Finally, create the Raw object
        raw = mne.io.RawArray(data_sub.T, info, verbose = True)

        # Filter

        raw.filter(1,30)

        # Set EEG reference
        raw.set_eeg_reference()
```

```

    # Plot it!
    raw.plot(scalings="auto", duration = 2)

    # Save it!
    if "Open" in file: # open and closed csv files have different
lengths, so by doing this we ensure the names of the new files will be
correct
        raw.save("2. Loading & Digitization/"+file[0:11]+"_raw.fif",
overwrite = True)
    else:
        raw.save("2. Loading & Digitization/"+file[0:13]+"_raw.fif",
overwrite = True)

```

Tokenization took: 16.57 ms

Type conversion took: 12.18 ms

Parser memory cleanup took: 0.00 ms

****audio.csv****

Creating RawArray with float64 data, n_channels=8, n_times=16288

Range : 0 ... 16287 = 0.000 ... 65.148 secs

Ready.

Filtering raw data in 1 contiguous segment

Setting up band-pass filter from 1 - 30 Hz

FIR filter parameters

Designing a one-pass, zero-phase, non-causal bandpass filter:

- Windowed time-domain design (firwin) method
- Hamming window with 0.0194 passband ripple and 53 dB stopband attenuation
- Lower passband edge: 1.00
- Lower transition bandwidth: 1.00 Hz (-6 dB cutoff frequency: 0.50 Hz)
- Upper passband edge: 30.00 Hz
- Upper transition bandwidth: 7.50 Hz (-6 dB cutoff frequency: 33.75 Hz)
- Filter length: 825 samples (3.300 s)

EEG channel type selected for re-referencing

Applying average reference.

Applying a custom ('EEG',) reference.

[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.

[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s

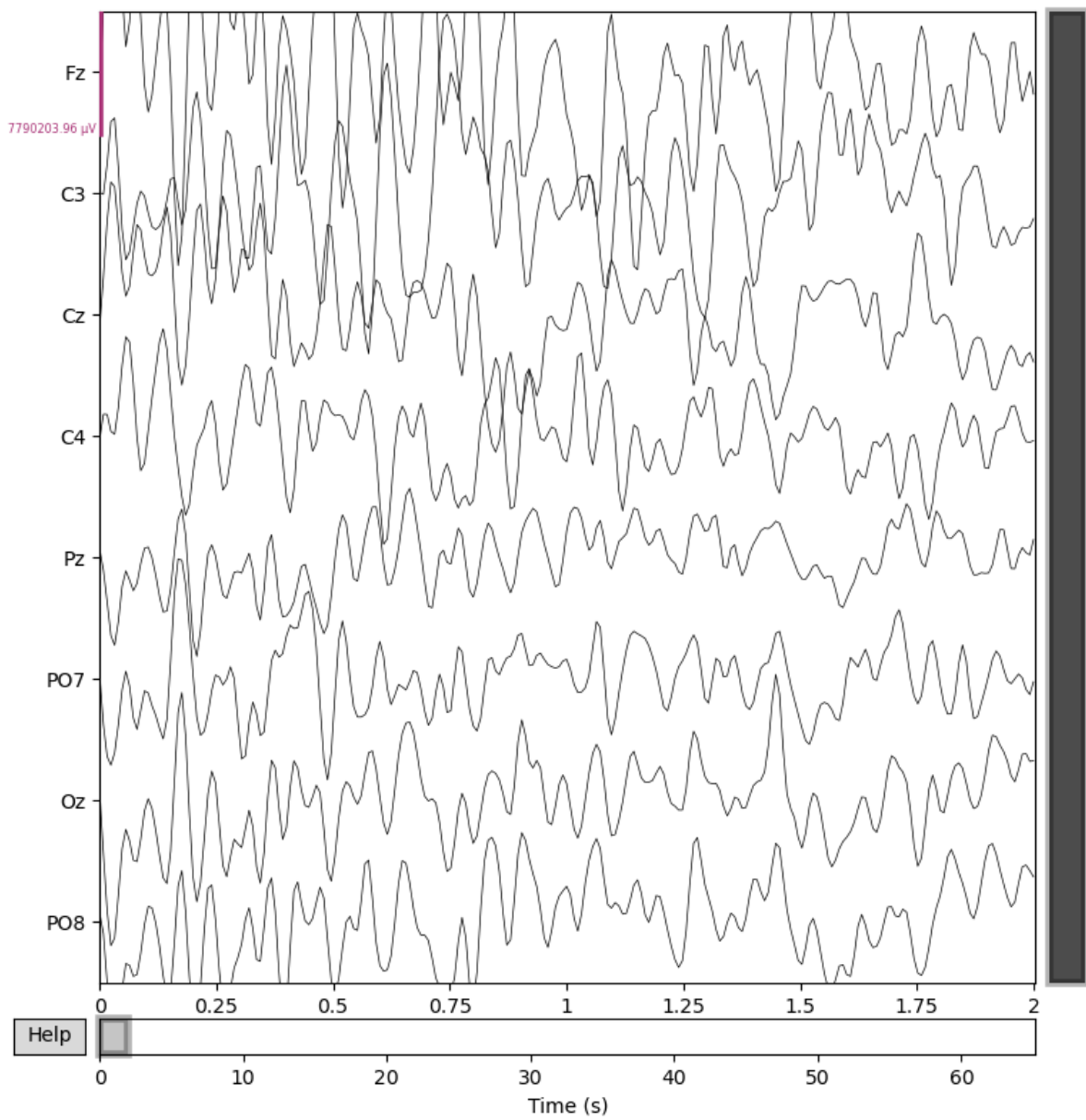
[Parallel(n_jobs=1)]: Done 2 out of 2 | elapsed: 0.0s remaining: 0.0s

[Parallel(n_jobs=1)]: Done 3 out of 3 | elapsed: 0.0s remaining: 0.0s

[Parallel(n_jobs=1)]: Done 4 out of 4 | elapsed: 0.0s remaining: 0.0s

[Parallel(n_jobs=1)]: Done 8 out of 8 | elapsed: 0.0s finished

Using matplotlib as 2D backend.

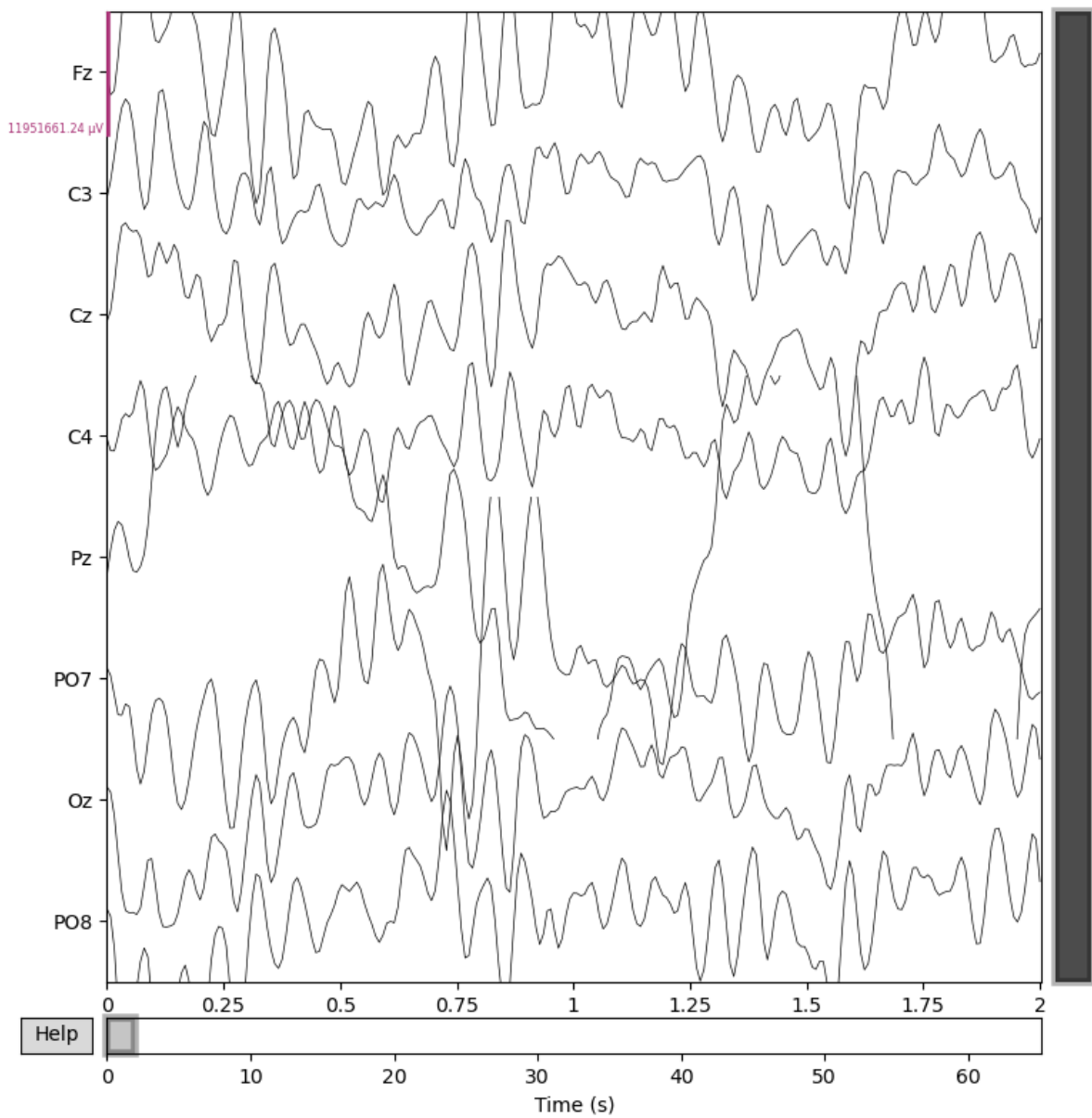


```
Overwriting existing file.
Writing C:\Users\IamAh\Desktop\UNIVERSITY - CSAI\Cognitive Neuroscience\M001_Cognitive_Neuroscience_Practical_Material\EEG_practical\2. Loading & Digitization\audio.csv_raw.fif
Closing C:\Users\IamAh\Desktop\UNIVERSITY - CSAI\Cognitive Neuroscience\M001_Cognitive_Neuroscience_Practical_Material\EEG_practical\2. Loading & Digitization\audio.csv_raw.fif
[done]
Tokenization took: 16.12 ms
Type conversion took: 8.50 ms
Parser memory cleanup took: 0.00 ms
****Closed_eyes_1.csv****
Creating RawArray with float64 data, n_channels=8, n_times=16280
    Range : 0 ... 16279 =      0.000 ...    65.116 secs
Ready.
Filtering raw data in 1 contiguous segment
Setting up band-pass filter from 1 - 30 Hz

FIR filter parameters
-----
Designing a one-pass, zero-phase, non-causal bandpass filter:
- Windowed time-domain design (firwin) method
- Hamming window with 0.0194 passband ripple and 53 dB stopband attenuation
- Lower passband edge: 1.00
- Lower transition bandwidth: 1.00 Hz (-6 dB cutoff frequency: 0.50 Hz)
- Upper passband edge: 30.00 Hz
- Upper transition bandwidth: 7.50 Hz (-6 dB cutoff frequency: 33.75 Hz)
- Filter length: 825 samples (3.300 s)

EEG channel type selected for re-referencing
Applying average reference.
Applying a custom ('EEG',) reference.

[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done   1 out of   1 | elapsed:   0.0s remaining:   0.0s
[Parallel(n_jobs=1)]: Done   2 out of   2 | elapsed:   0.0s remaining:   0.0s
[Parallel(n_jobs=1)]: Done   3 out of   3 | elapsed:   0.0s remaining:   0.0s
[Parallel(n_jobs=1)]: Done   4 out of   4 | elapsed:   0.0s remaining:   0.0s
[Parallel(n_jobs=1)]: Done   8 out of   8 | elapsed:   0.0s finished
```

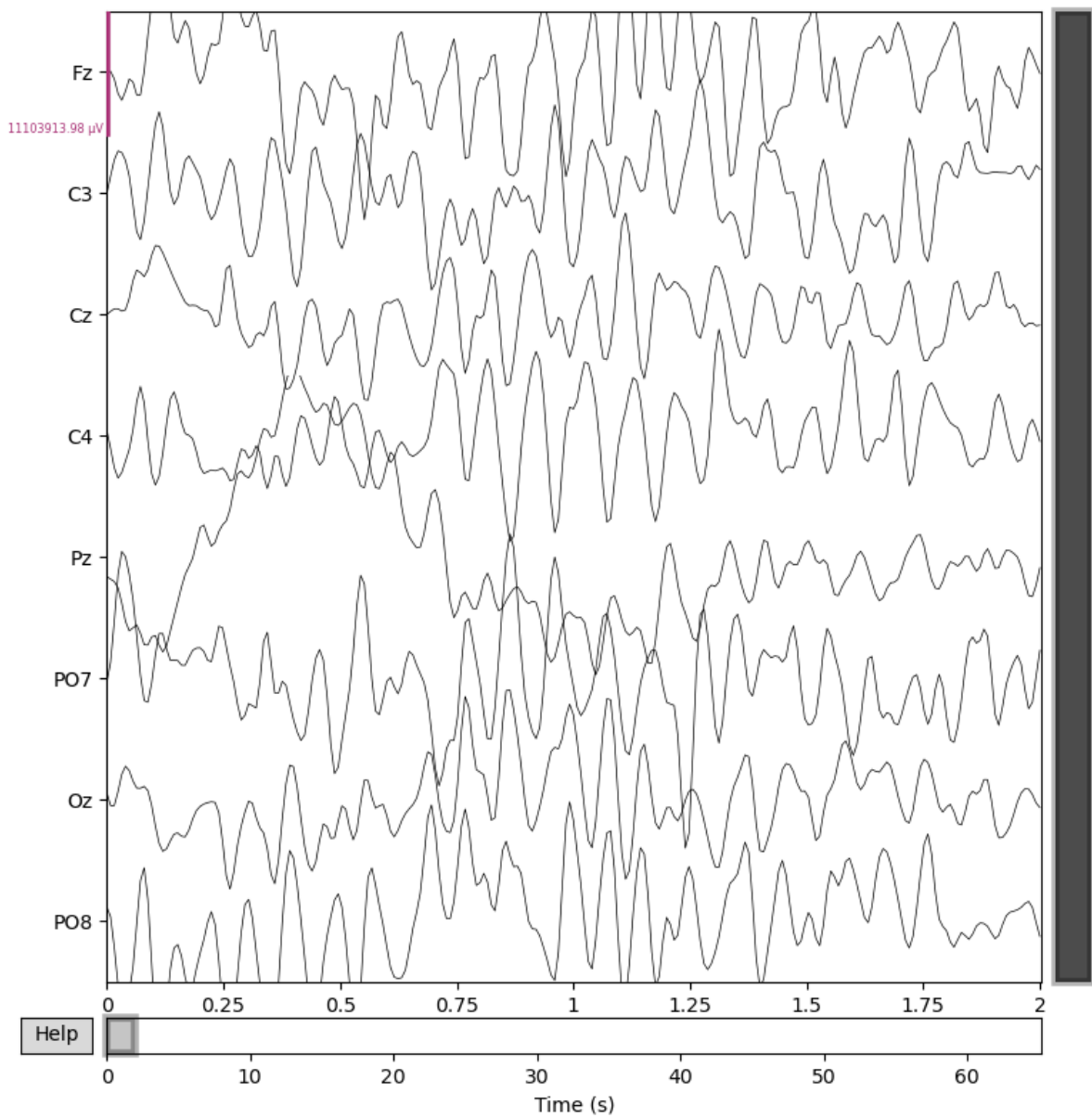


```
Overwriting existing file.
Writing C:\Users\IamAh\Desktop\UNIVERSITY - CSAI\Cognitive Neuroscience\M001_Cognitive_Neuroscience_Practical_Material\EEG_practical\2. Loading & Digitization\Closed_eyes_1_raw.fif
Closing C:\Users\IamAh\Desktop\UNIVERSITY - CSAI\Cognitive Neuroscience\M001_Cognitive_Neuroscience_Practical_Material\EEG_practical\2. Loading & Digitization\Closed_eyes_1_raw.fif
[done]
Tokenization took: 19.58 ms
Type conversion took: 8.06 ms
Parser memory cleanup took: 0.00 ms
****Closed_eyes_2.csv****
Creating RawArray with float64 data, n_channels=8, n_times=16304
    Range : 0 ... 16303 =      0.000 ...    65.212 secs
Ready.
Filtering raw data in 1 contiguous segment
Setting up band-pass filter from 1 - 30 Hz

FIR filter parameters
-----
Designing a one-pass, zero-phase, non-causal bandpass filter:
- Windowed time-domain design (firwin) method
- Hamming window with 0.0194 passband ripple and 53 dB stopband attenuation
- Lower passband edge: 1.00
- Lower transition bandwidth: 1.00 Hz (-6 dB cutoff frequency: 0.50 Hz)
- Upper passband edge: 30.00 Hz
- Upper transition bandwidth: 7.50 Hz (-6 dB cutoff frequency: 33.75 Hz)
- Filter length: 825 samples (3.300 s)

EEG channel type selected for re-referencing
Applying average reference.
Applying a custom ('EEG',) reference.

[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done   1 out of   1 | elapsed:    0.0s remaining:    0.0s
[Parallel(n_jobs=1)]: Done   2 out of   2 | elapsed:    0.0s remaining:    0.0s
[Parallel(n_jobs=1)]: Done   3 out of   3 | elapsed:    0.0s remaining:    0.0s
[Parallel(n_jobs=1)]: Done   4 out of   4 | elapsed:    0.0s remaining:    0.0s
[Parallel(n_jobs=1)]: Done   8 out of   8 | elapsed:    0.0s finished
```

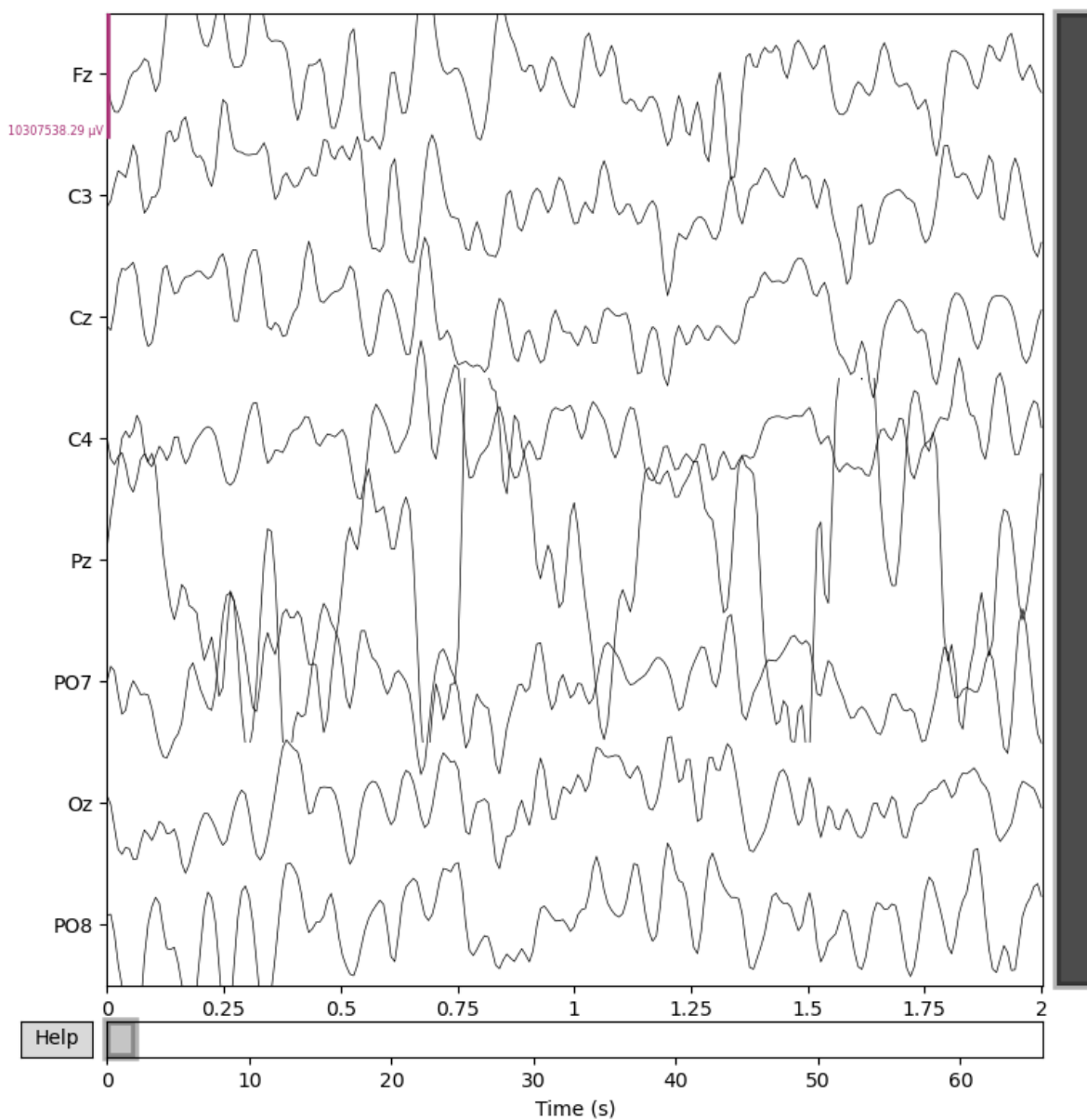


```
Overwriting existing file.
Writing C:\Users\IamAh\Desktop\UNIVERSITY - CSAI\Cognitive Neuroscience\M001_Cognitive_Neuroscience_Practical_Material\EEG_practical\2. Loading & Digitization\Closed_eyes_2_raw.fif
Closing C:\Users\IamAh\Desktop\UNIVERSITY - CSAI\Cognitive Neuroscience\M001_Cognitive_Neuroscience_Practical_Material\EEG_practical\2. Loading & Digitization\Closed_eyes_2_raw.fif
[done]
Tokenization took: 20.11 ms
Type conversion took: 12.35 ms
Parser memory cleanup took: 0.00 ms
****Open_eyes_1.csv****
Creating RawArray with float64 data, n_channels=8, n_times=16456
    Range : 0 ... 16455 =      0.000 ...    65.820 secs
Ready.
Filtering raw data in 1 contiguous segment
Setting up band-pass filter from 1 - 30 Hz

FIR filter parameters
-----
Designing a one-pass, zero-phase, non-causal bandpass filter:
- Windowed time-domain design (firwin) method
- Hamming window with 0.0194 passband ripple and 53 dB stopband attenuation
- Lower passband edge: 1.00
- Lower transition bandwidth: 1.00 Hz (-6 dB cutoff frequency: 0.50 Hz)
- Upper passband edge: 30.00 Hz
- Upper transition bandwidth: 7.50 Hz (-6 dB cutoff frequency: 33.75 Hz)
- Filter length: 825 samples (3.300 s)

EEG channel type selected for re-referencing
Applying average reference.
Applying a custom ('EEG',) reference.

[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done   1 out of   1 | elapsed:   0.0s remaining:   0.0s
[Parallel(n_jobs=1)]: Done   2 out of   2 | elapsed:   0.0s remaining:   0.0s
[Parallel(n_jobs=1)]: Done   3 out of   3 | elapsed:   0.0s remaining:   0.0s
[Parallel(n_jobs=1)]: Done   4 out of   4 | elapsed:   0.0s remaining:   0.0s
[Parallel(n_jobs=1)]: Done   8 out of   8 | elapsed:   0.0s finished
```

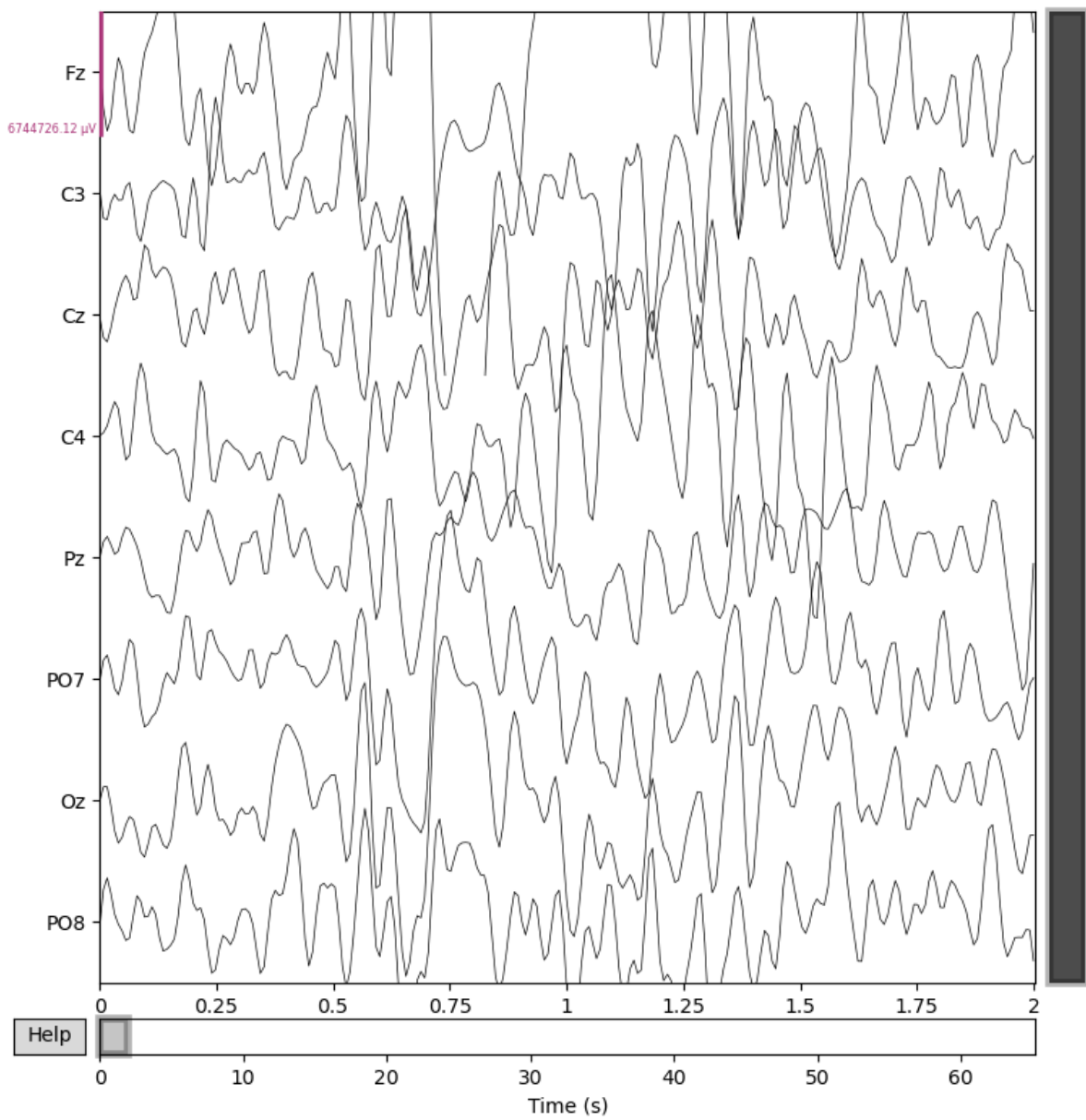


```
Overwriting existing file.
Writing C:\Users\IamAh\Desktop\UNIVERSITY - CSAI\Cognitive Neuroscience\M001_Cognitive_Neuroscience_Practical_Material\EEG_practical\2. Loading & Digitization\Open_eyes_1_raw.fif
Closing C:\Users\IamAh\Desktop\UNIVERSITY - CSAI\Cognitive Neuroscience\M001_Cognitive_Neuroscience_Practical_Material\EEG_practical\2. Loading & Digitization\Open_eyes_1_raw.fif
[done]
Tokenization took: 16.17 ms
Type conversion took: 12.61 ms
Parser memory cleanup took: 0.00 ms
****Open_eyes_2.csv****
Creating RawArray with float64 data, n_channels=8, n_times=16296
    Range : 0 ... 16295 =      0.000 ...    65.180 secs
Ready.
Filtering raw data in 1 contiguous segment
Setting up band-pass filter from 1 - 30 Hz

FIR filter parameters
-----
Designing a one-pass, zero-phase, non-causal bandpass filter:
- Windowed time-domain design (firwin) method
- Hamming window with 0.0194 passband ripple and 53 dB stopband attenuation
- Lower passband edge: 1.00
- Lower transition bandwidth: 1.00 Hz (-6 dB cutoff frequency: 0.50 Hz)
- Upper passband edge: 30.00 Hz
- Upper transition bandwidth: 7.50 Hz (-6 dB cutoff frequency: 33.75 Hz)
- Filter length: 825 samples (3.300 s)

EEG channel type selected for re-referencing
Applying average reference.
Applying a custom ('EEG',) reference.

[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done   1 out of   1 | elapsed:   0.0s remaining:   0.0s
[Parallel(n_jobs=1)]: Done   2 out of   2 | elapsed:   0.0s remaining:   0.0s
[Parallel(n_jobs=1)]: Done   3 out of   3 | elapsed:   0.0s remaining:   0.0s
[Parallel(n_jobs=1)]: Done   4 out of   4 | elapsed:   0.0s remaining:   0.0s
[Parallel(n_jobs=1)]: Done   8 out of   8 | elapsed:   0.0s finished
```

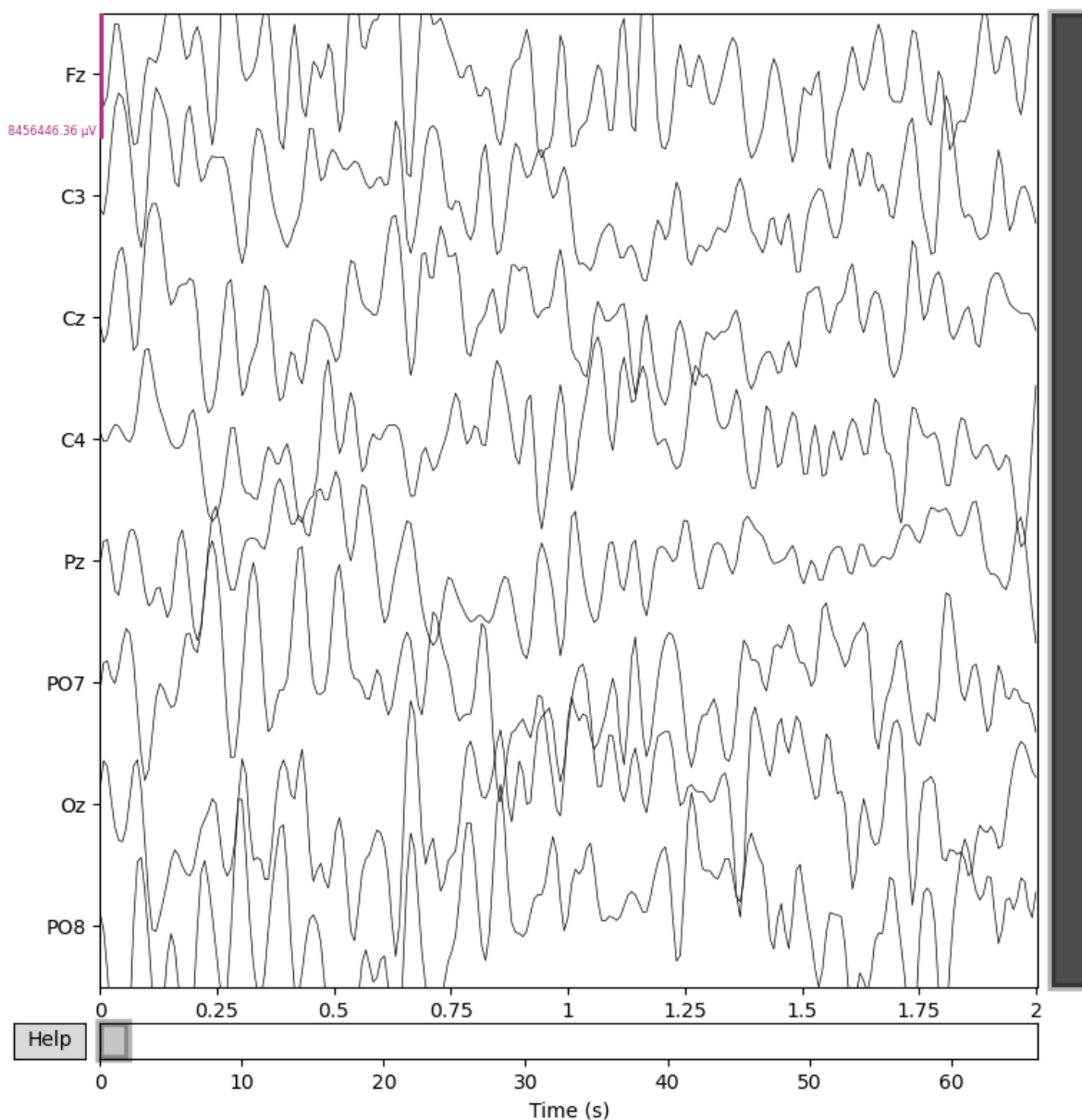


```
Overwriting existing file.
Writing C:\Users\IamAh\Desktop\UNIVERSITY - CSAI\Cognitive Neuroscience\M001_Cognitive_Neuroscience_Practical_Material\EEG_practical\2. Loading & Digitization\Open_eyes_2_raw.fif
Closing C:\Users\IamAh\Desktop\UNIVERSITY - CSAI\Cognitive Neuroscience\M001_Cognitive_Neuroscience_Practical_Material\EEG_practical\2. Loading & Digitization\Open_eyes_2_raw.fif
[done]
Tokenization took: 14.09 ms
Type conversion took: 10.00 ms
Parser memory cleanup took: 0.00 ms
****vision.csv****
Creating RawArray with float64 data, n_channels=8, n_times=16528
    Range : 0 ... 16527 =      0.000 ...    66.108 secs
Ready.
Filtering raw data in 1 contiguous segment
Setting up band-pass filter from 1 - 30 Hz

FIR filter parameters
-----
Designing a one-pass, zero-phase, non-causal bandpass filter:
- Windowed time-domain design (firwin) method
- Hamming window with 0.0194 passband ripple and 53 dB stopband attenuation
- Lower passband edge: 1.00
- Lower transition bandwidth: 1.00 Hz (-6 dB cutoff frequency: 0.50 Hz)
- Upper passband edge: 30.00 Hz
- Upper transition bandwidth: 7.50 Hz (-6 dB cutoff frequency: 33.75 Hz)
- Filter length: 825 samples (3.300 s)

EEG channel type selected for re-referencing
Applying average reference.
Applying a custom ('EEG',) reference.

[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done   1 out of   1 | elapsed:   0.0s remaining:   0.0s
[Parallel(n_jobs=1)]: Done   2 out of   2 | elapsed:   0.0s remaining:   0.0s
[Parallel(n_jobs=1)]: Done   3 out of   3 | elapsed:   0.0s remaining:   0.0s
[Parallel(n_jobs=1)]: Done   4 out of   4 | elapsed:   0.0s remaining:   0.0s
[Parallel(n_jobs=1)]: Done   8 out of   8 | elapsed:   0.0s finished
```



Overwriting existing file.

Writing C:\Users\IamAh\Desktop\UNIVERSITY - CSAI\Cognitive Neuroscience\M001_Cognitive_Neuroscience_Practical_Material\EEG_practical\2. Loading & Digitization\vision.csv_raw.fif

Closing C:\Users\IamAh\Desktop\UNIVERSITY - CSAI\Cognitive Neuroscience\M001_Cognitive_Neuroscience_Practical_Material\EEG_practical\2. Loading & Digitization\vision.csv_raw.fif

[done]

For .vhdr extension - skip this

```
In [8]: """for eo_vhdr_file in os.listdir("1. Raw files"):
        if "Open" in eo_vhdr_file:
            sample_data_raw_file = ("1. Raw files/"+eo_vhdr_file)
            raw = mne.io.read_raw_brainvision(sample_data_raw_file)
            raw.plot_psd(fmax=50, verbose = False)
            raw.save("2. Loading & Digitization/"+file[0:11]+"_raw.fif")"""
```

```
Out[8]: 'for eo_vhdr_file in os.listdir("1. Raw files"):
        if "Open" in eo_vhdr_file:
            sample_data_raw_file = ("1. Raw files/"+eo_vhdr_file)
            raw = mne.io.read_raw_brainvision(sample_data_raw_file)
            raw.plot_psd(fmax=50, verbose = False)
            raw.save("2. Loading & Digitization/"+file[0:11]+"_raw.fif")'
```

```
In [9]: """for ec_vhdr_file in os.listdir("1. Raw files"):
        if "Closed" in ec_vhdr_file:
            sample_data_raw_file = ("1. Raw files/"+ec_vhdr_file)
            raw = mne.io.read_raw_brainvision(sample_data_raw_file)
            raw.plot_psd(fmax=50, verbose = False)
            raw.save("2. Loading & Digitization/"+file[0:13]+"_raw.fif")"""
```

```
Out[9]: 'for ec_vhdr_file in os.listdir("1. Raw files"):
        if "Closed" in ec_vhdr_file:
            sample_data_raw_file = ("1. Raw files/"+ec_vhdr_file)
            raw = mne.io.read_raw_brainvision(sample_data_raw_file)
            raw.plot_psd(fmax=50, verbose = False)
            raw.save("2. Loading & Digitization/"+file[0:13]+"_raw.fif")'
```

Step 3: artifacts rejection

Artifacts rejection is an important step in the EEG data analysis process: it allows to remove unwanted noise in the signal, possibly caused by (among others):

1. eye blinks
2. saccades
3. muscle twitches
4. noise in the environment
5. electrical signal sources in proximity (i.e., laptops, EEG hardware etc.)

For this tutorial, we will focus on eye blinks and saccades, the two most prominent artifacts.

It's important to know that the existing algorithms that are meant to target and remove these type of artifacts are great, but occasionally still report False Negatives (an artifact is classified as clean signal) as well as False Positives (a clean signal is classified as artifact).

Run the following code to perform automatic eye artifacts rejection:

```
In [10]: from ica_function import ica_visualization

path_to_dig_file = ["2. Loading & Digitization"] # define the path to
the digitized file
```

```
In [11]: for i in range(len(path_to_dig_file)):
    files_list = os.listdir(path_to_dig_file[i])
    for filename in files_list: # for file in files_list
        if "raw" in filename: # if this file has been digitized
            ica_visualization(filename, path_to_dig_file[i]) # run
automatic ICA rejection
```

Opening raw data file 2. Loading & Digitization/audio.csv_raw.fif...

Range : 0 ... 16287 = 0.000 ... 65.148 secs

Ready.

Reading 0 ... 16287 = 0.000 ... 65.148 secs...

Filtering raw data in 1 contiguous segment

Setting up band-pass filter from 1 - 1e+02 Hz

FIR filter parameters

Designing a one-pass, zero-phase, non-causal bandpass filter:

- Windowed time-domain design (firwin) method
- Hamming window with 0.0194 passband ripple and 53 dB stopband attenuation
- Lower passband edge: 1.00
- Lower transition bandwidth: 1.00 Hz (-6 dB cutoff frequency: 0.50 Hz)
- Upper passband edge: 100.00 Hz
- Upper transition bandwidth: 25.00 Hz (-6 dB cutoff frequency: 112.50 Hz)
- Filter length: 825 samples (3.300 s)

Fitting ICA to data using 7 channels (please be patient, this may take a while)

Selecting by number: 7 components

[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.

[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s

[Parallel(n_jobs=1)]: Done 2 out of 2 | elapsed: 0.0s remaining: 0.0s

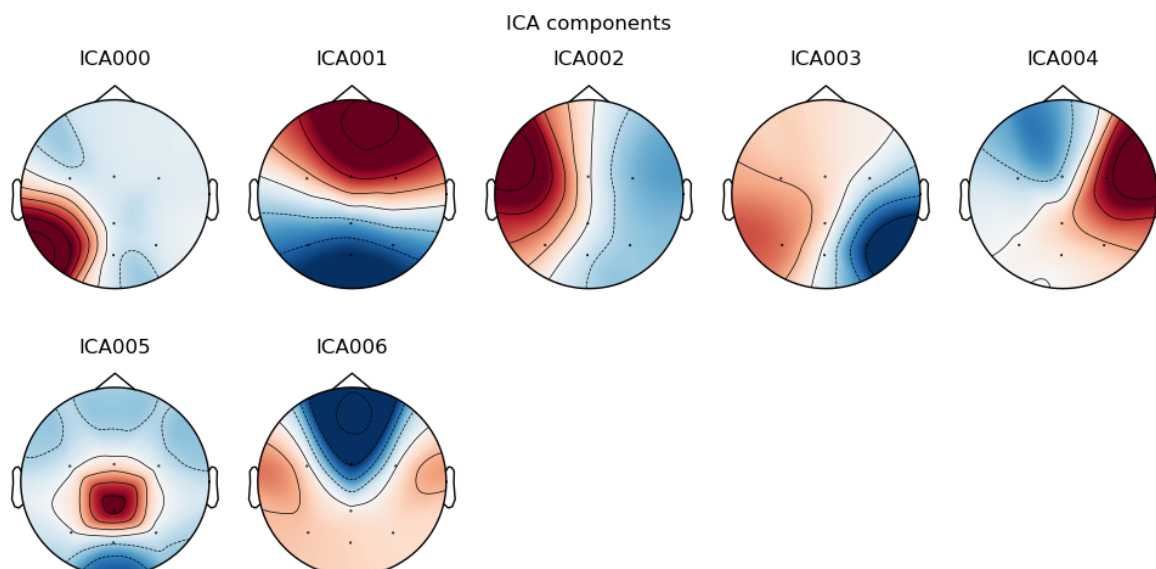
[Parallel(n_jobs=1)]: Done 3 out of 3 | elapsed: 0.0s remaining: 0.0s

[Parallel(n_jobs=1)]: Done 4 out of 4 | elapsed: 0.0s remaining: 0.0s

[Parallel(n_jobs=1)]: Done 8 out of 8 | elapsed: 0.0s finished

Fitting ICA took 0.2s.

****No ICA components to remove found****



Applying ICA to Raw instance

Transforming to ICA space (7 components)

Zeroing out 0 ICA components

Projecting back using 7 PCA components

Writing C:\Users\IamAh\Desktop\UNIVERSITY - CSAI\Cognitive Neuroscience\M001_Cognitive_Neuroscience_Practical_Material\EEG_practical\3. ICA\audio.csv_r_ica.fif

Closing C:\Users\IamAh\Desktop\UNIVERSITY - CSAI\Cognitive Neuroscience\M001_Cognitive_Neuroscience_Practical_Material\EEG_practical\3. ICA\audio.csv_r_ica.fif

[done]

Opening raw data file 2. Loading & Digitization/Closed_eyes_1_raw.fif...

Range : 0 ... 16279 = 0.000 ... 65.116 secs

Ready.

Reading 0 ... 16279 = 0.000 ... 65.116 secs...

Filtering raw data in 1 contiguous segment

Setting up band-pass filter from 1 - 1e+02 Hz

FIR filter parameters

Designing a one-pass, zero-phase, non-causal bandpass filter:

- Windowed time-domain design (firwin) method
- Hamming window with 0.0194 passband ripple and 53 dB stopband attenuation
- Lower passband edge: 1.00
- Lower transition bandwidth: 1.00 Hz (-6 dB cutoff frequency: 0.50 Hz)
- Upper passband edge: 100.00 Hz
- Upper transition bandwidth: 25.00 Hz (-6 dB cutoff frequency: 112.50 Hz)
- Filter length: 825 samples (3.300 s)

Fitting ICA to data using 7 channels (please be patient, this may take a while)

Selecting by number: 7 components

Fitting ICA took 0.0s.

****No ICA components to remove found****

```
C:\Users\IamAh\Desktop\UNIVERSITY - CSAI\Cognitive Neuroscience\M001_Cognitive_Neuroscience_Practical_Material\EEG_practical\ica_function.py:45: RuntimeWarning: This filename (C:\Users\IamAh\Desktop\UNIVERSITY - CSAI\Cognitive Neuroscience\M001_Cognitive_Neuroscience_Practical_Material\EEG_practical\3. ICA\audio.csv_r_ica.fif) does not conform to MNE naming conventions. All raw files should end with raw.fif, raw_sss.fif, raw_tsss.fif, _meg.fif, _eeg.fif, _ieeg.fif, raw.fif.gz, raw_sss.fif.gz, raw_tsss.fif.gz, _meg.fif.gz, _eeg.fif.gz or _ieeg.fif.gz
```

```
raw.save(fname = "3. ICA/"+new_name, overwrite = True)
```

```
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
```

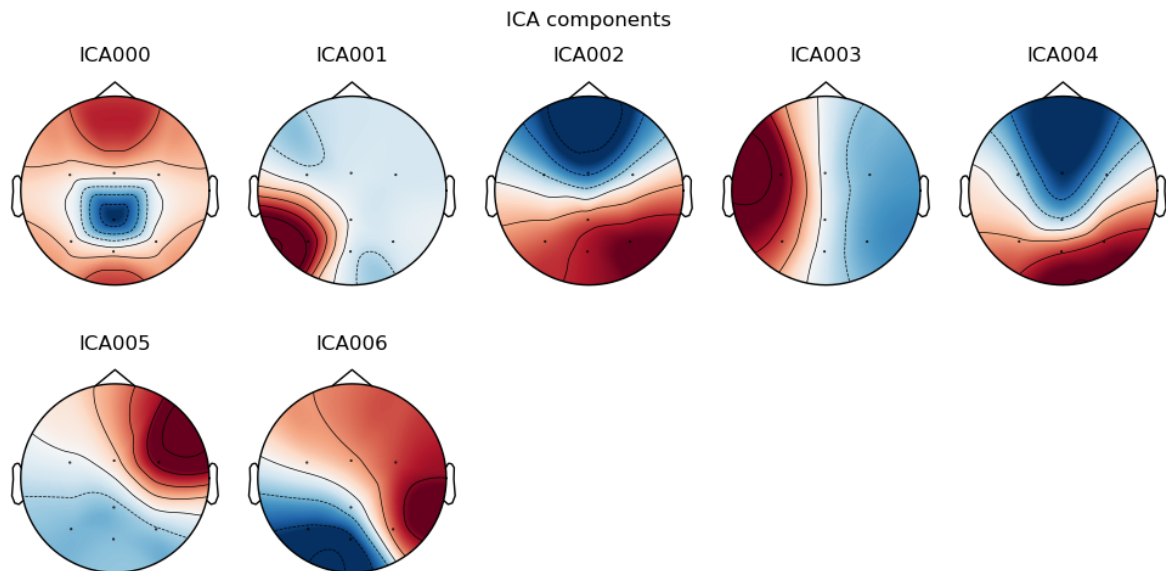
```
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
```

```
[Parallel(n_jobs=1)]: Done 2 out of 2 | elapsed: 0.0s remaining: 0.0s
```

```
[Parallel(n_jobs=1)]: Done 3 out of 3 | elapsed: 0.0s remaining: 0.0s
```

```
[Parallel(n_jobs=1)]: Done 4 out of 4 | elapsed: 0.0s remaining: 0.0s
```

```
[Parallel(n_jobs=1)]: Done 8 out of 8 | elapsed: 0.0s finished
```



```

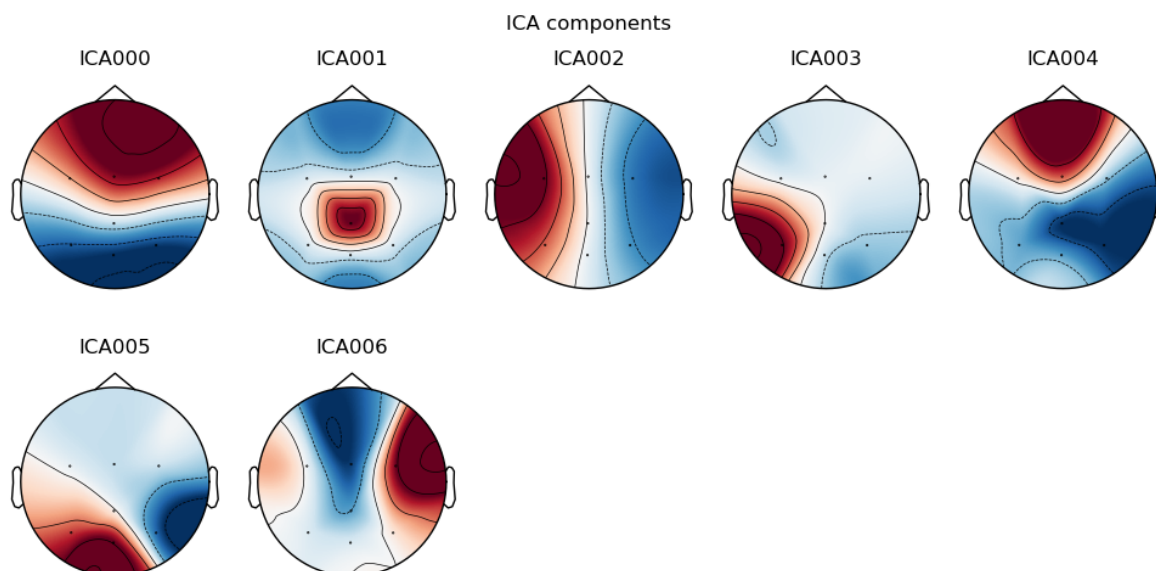
Applying ICA to Raw instance
    Transforming to ICA space (7 components)
    Zeroing out 0 ICA components
    Projecting back using 7 PCA components
Writing C:\Users\IamAh\Desktop\UNIVERSITY - CSAI\Cognitive Neuroscience\M001_Cognitive_Neuroscience_Practical_Material\EEG_practical\3. ICA\Closed_eyes_1_ica.fif
Closing C:\Users\IamAh\Desktop\UNIVERSITY - CSAI\Cognitive Neuroscience\M001_Cognitive_Neuroscience_Practical_Material\EEG_practical\3. ICA\Closed_eyes_1_ica.fif
[done]
Opening raw data file 2. Loading & Digitization/Closed_eyes_2_raw.fif...
    Range : 0 ... 16303 =      0.000 ...    65.212 secs
Ready.
Reading 0 ... 16303 =      0.000 ...    65.212 secs...
Filtering raw data in 1 contiguous segment
Setting up band-pass filter from 1 - 1e+02 Hz

FIR filter parameters
-----
Designing a one-pass, zero-phase, non-causal bandpass filter:
- Windowed time-domain design (firwin) method
- Hamming window with 0.0194 passband ripple and 53 dB stopband attenuation
- Lower passband edge: 1.00
- Lower transition bandwidth: 1.00 Hz (-6 dB cutoff frequency: 0.50 Hz)
- Upper passband edge: 100.00 Hz
- Upper transition bandwidth: 25.00 Hz (-6 dB cutoff frequency: 112.50 Hz)
- Filter length: 825 samples (3.300 s)

Fitting ICA to data using 7 channels (please be patient, this may take a while)
Selecting by number: 7 components
Fitting ICA took 0.0s.
****No ICA components to remove found****

C:\Users\IamAh\Desktop\UNIVERSITY - CSAI\Cognitive Neuroscience\M001_Cognitive_Neuroscience_Practical_Material\EEG_practical\ica_function.py:42: RuntimeWarning: This filename (C:\Users\IamAh\Desktop\UNIVERSITY - CSAI\Cognitive Neuroscience\M001_Cognitive_Neuroscience_Practical_Material\EEG_practical\3. ICA\Closed_eyes_1_ica.fif) does not conform to MNE naming conventions. All raw files should end with raw.fif, raw_sss.fif, raw_tsss.fif, _meg.fif, _eeg.fif, _ieeg.fif, raw.fif.gz, raw_sss.fif.gz, raw_tsss.fif.gz, _meg.fif.gz, _eeg.fif.gz or _ieeg.fif.gz
  raw.save(fname = "3. ICA/"+new_name, overwrite = True)
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done   1 out of   1 | elapsed:    0.0s remaining:    0.0s
[Parallel(n_jobs=1)]: Done   2 out of   2 | elapsed:    0.0s remaining:    0.0s
[Parallel(n_jobs=1)]: Done   3 out of   3 | elapsed:    0.0s remaining:    0.0s
[Parallel(n_jobs=1)]: Done   4 out of   4 | elapsed:    0.0s remaining:    0.0s
[Parallel(n_jobs=1)]: Done   8 out of   8 | elapsed:    0.0s finished

```



Applying ICA to Raw instance

Transforming to ICA space (7 components)

Zeroing out 0 ICA components

Projecting back using 7 PCA components

Writing C:\Users\IamAh\Desktop\UNIVERSITY - CSAI\Cognitive Neuroscience\M001_Cognitive_Neuroscience_Practical_Material\EEG_practical\3. ICA\Closed_eyes_2_ica.fif

Closing C:\Users\IamAh\Desktop\UNIVERSITY - CSAI\Cognitive Neuroscience\M001_Cognitive_Neuroscience_Practical_Material\EEG_practical\3. ICA\Closed_eyes_2_ica.fif

[done]

Opening raw data file 2. Loading & Digitization/Open_eyes_1_raw.fif...

Range : 0 ... 16455 = 0.000 ... 65.820 secs

Ready.

Reading 0 ... 16455 = 0.000 ... 65.820 secs...

Filtering raw data in 1 contiguous segment

Setting up band-pass filter from 1 - 1e+02 Hz

FIR filter parameters

Designing a one-pass, zero-phase, non-causal bandpass filter:

- Windowed time-domain design (firwin) method
- Hamming window with 0.0194 passband ripple and 53 dB stopband attenuation
- Lower passband edge: 1.00
- Lower transition bandwidth: 1.00 Hz (-6 dB cutoff frequency: 0.50 Hz)
- Upper passband edge: 100.00 Hz
- Upper transition bandwidth: 25.00 Hz (-6 dB cutoff frequency: 112.50 Hz)
- Filter length: 825 samples (3.300 s)

Fitting ICA to data using 7 channels (please be patient, this may take a while)

Selecting by number: 7 components

Fitting ICA took 0.0s.

****No ICA components to remove found****

```
C:\Users\IamAh\Desktop\UNIVERSITY - CSAI\Cognitive Neuroscience\M001_Cognitive_Neuroscience_Practical_Material\EEG_practical\ica_function.py:42: RuntimeWarning: This filename (C:\Users\IamAh\Desktop\UNIVERSITY - CSAI\Cognitive Neuroscience\M001_Cognitive_Neuroscience_Practical_Material\EEG_practical\3. ICA\Closed_eyes_2_ica.fif) does not conform to MNE naming conventions. All raw files should end with raw.fif, raw_sss.fif, raw_tsss.fif, _meg.fif, _eeg.fif, _ieeg.fif, raw.fif.gz, raw_sss.fif.gz, raw_tsss.fif.gz, _meg.fif.gz, _eeg.fif.gz or _ieeg.fif.gz
```

```
raw.save(fname = "3. ICA/"+new_name, overwrite = True)
```

```
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
```

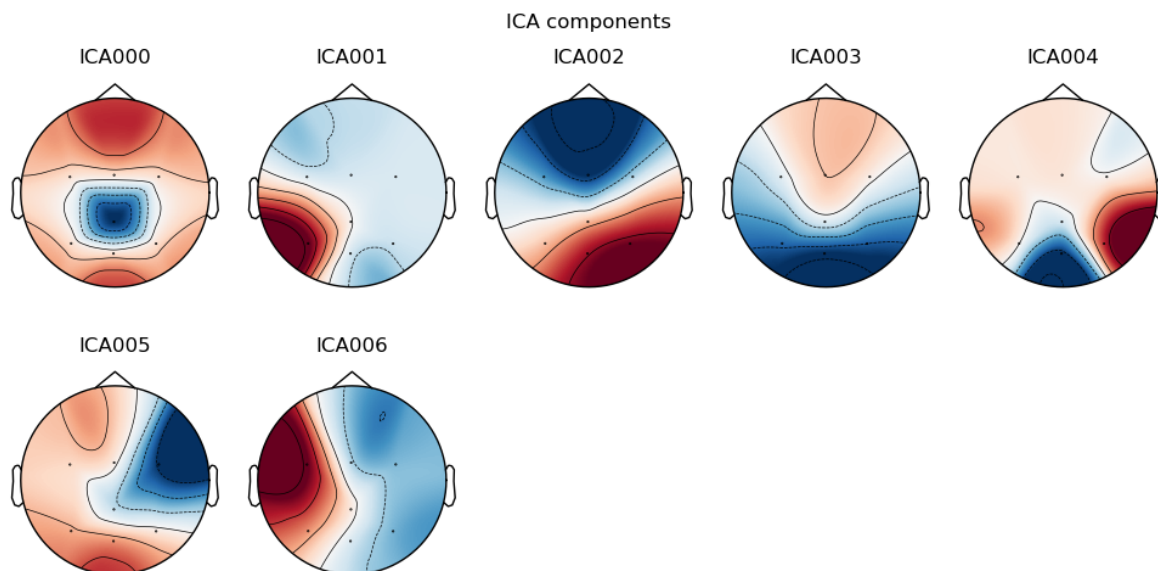
```
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
```

```
[Parallel(n_jobs=1)]: Done 2 out of 2 | elapsed: 0.0s remaining: 0.0s
```

```
[Parallel(n_jobs=1)]: Done 3 out of 3 | elapsed: 0.0s remaining: 0.0s
```

```
[Parallel(n_jobs=1)]: Done 4 out of 4 | elapsed: 0.0s remaining: 0.0s
```

```
[Parallel(n_jobs=1)]: Done 8 out of 8 | elapsed: 0.0s finished
```



```

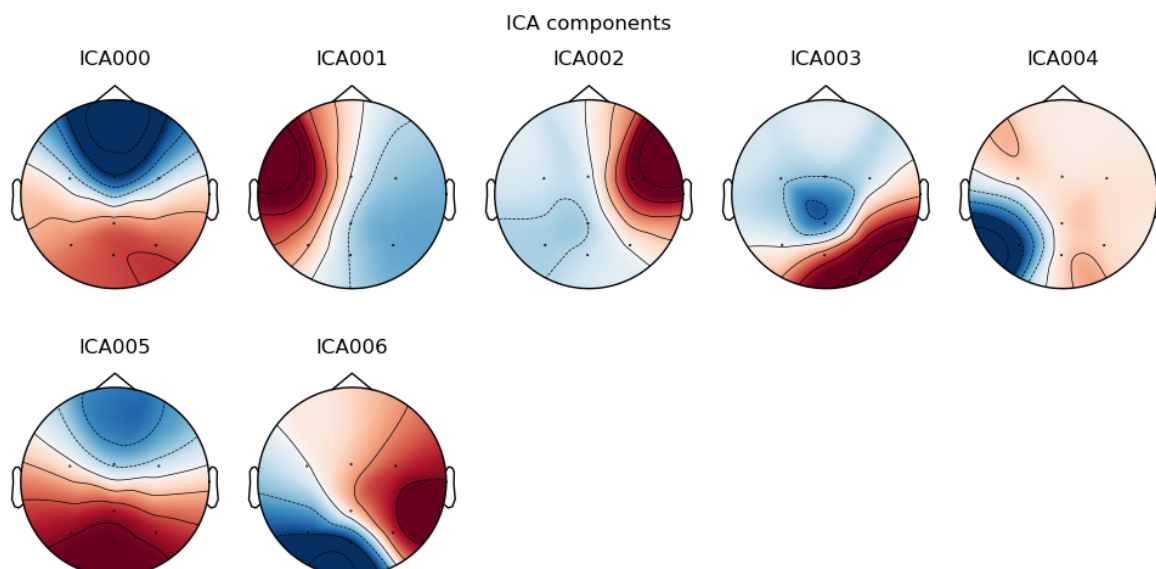
Applying ICA to Raw instance
    Transforming to ICA space (7 components)
    Zeroing out 0 ICA components
    Projecting back using 7 PCA components
Writing C:\Users\IamAh\Desktop\UNIVERSITY - CSAI\Cognitive Neuroscience\M001_Cognitive_Neuroscience_Practical_Material\EEG_practical\3. ICA\Open_eyes_1_ica.fif
Closing C:\Users\IamAh\Desktop\UNIVERSITY - CSAI\Cognitive Neuroscience\M001_Cognitive_Neuroscience_Practical_Material\EEG_practical\3. ICA\Open_eyes_1_ica.fif
[done]
Opening raw data file 2. Loading & Digitization/Open_eyes_2_raw.fif...
    Range : 0 ... 16295 =      0.000 ...    65.180 secs
Ready.
Reading 0 ... 16295 =      0.000 ...    65.180 secs...
Filtering raw data in 1 contiguous segment
Setting up band-pass filter from 1 - 1e+02 Hz

FIR filter parameters
-----
Designing a one-pass, zero-phase, non-causal bandpass filter:
- Windowed time-domain design (firwin) method
- Hamming window with 0.0194 passband ripple and 53 dB stopband attenuation
- Lower passband edge: 1.00
- Lower transition bandwidth: 1.00 Hz (-6 dB cutoff frequency: 0.50 Hz)
- Upper passband edge: 100.00 Hz
- Upper transition bandwidth: 25.00 Hz (-6 dB cutoff frequency: 112.50 Hz)
- Filter length: 825 samples (3.300 s)

Fitting ICA to data using 7 channels (please be patient, this may take a while)
Selecting by number: 7 components
Fitting ICA took 0.0s.
****No ICA components to remove found****

C:\Users\IamAh\Desktop\UNIVERSITY - CSAI\Cognitive Neuroscience\M001_Cognitive_Neuroscience_Practical_Material\EEG_practical\ica_function.py:45: RuntimeWarning: This filename (C:\Users\IamAh\Desktop\UNIVERSITY - CSAI\Cognitive Neuroscience\M001_Cognitive_Neuroscience_Practical_Material\EEG_practical\3. ICA\Open_eyes_1_ica.fif) does not conform to MNE naming conventions. All raw files should end with raw.fif, raw_sss.fif, raw_tsss.fif, _meg.fif, _eeg.fif, _ieeg.fif, raw.fif.gz, raw_sss.fif.gz, raw_tsss.fif.gz, _meg.fif.gz, _eeg.fif.gz or _ieeg.fif.gz
  raw.save(fname = "3. ICA/"+new_name, overwrite = True)
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done   1 out of   1 | elapsed:    0.0s remaining:    0.0s
[Parallel(n_jobs=1)]: Done   2 out of   2 | elapsed:    0.0s remaining:    0.0s
[Parallel(n_jobs=1)]: Done   3 out of   3 | elapsed:    0.0s remaining:    0.0s
[Parallel(n_jobs=1)]: Done   4 out of   4 | elapsed:    0.0s remaining:    0.0s
[Parallel(n_jobs=1)]: Done   8 out of   8 | elapsed:    0.0s finished

```



Applying ICA to Raw instance

Transforming to ICA space (7 components)

Zeroing out 0 ICA components

Projecting back using 7 PCA components

Writing C:\Users\IamAh\Desktop\UNIVERSITY - CSAI\Cognitive Neuroscience\M001_Cognitive_Neuroscience_Practical_Material\EEG_practical\3. ICA\Open_eyes_2_ica.fif

Closing C:\Users\IamAh\Desktop\UNIVERSITY - CSAI\Cognitive Neuroscience\M001_Cognitive_Neuroscience_Practical_Material\EEG_practical\3. ICA\Open_eyes_2_ica.fif

[done]

Opening raw data file 2. Loading & Digitization/vision.csv_raw.fif...

Range : 0 ... 16527 = 0.000 ... 66.108 secs

Ready.

Reading 0 ... 16527 = 0.000 ... 66.108 secs...

Filtering raw data in 1 contiguous segment

Setting up band-pass filter from 1 - 1e+02 Hz

FIR filter parameters

Designing a one-pass, zero-phase, non-causal bandpass filter:

- Windowed time-domain design (firwin) method
- Hamming window with 0.0194 passband ripple and 53 dB stopband attenuation
- Lower passband edge: 1.00
- Lower transition bandwidth: 1.00 Hz (-6 dB cutoff frequency: 0.50 Hz)
- Upper passband edge: 100.00 Hz
- Upper transition bandwidth: 25.00 Hz (-6 dB cutoff frequency: 112.50 Hz)
- Filter length: 825 samples (3.300 s)

Fitting ICA to data using 7 channels (please be patient, this may take a while)

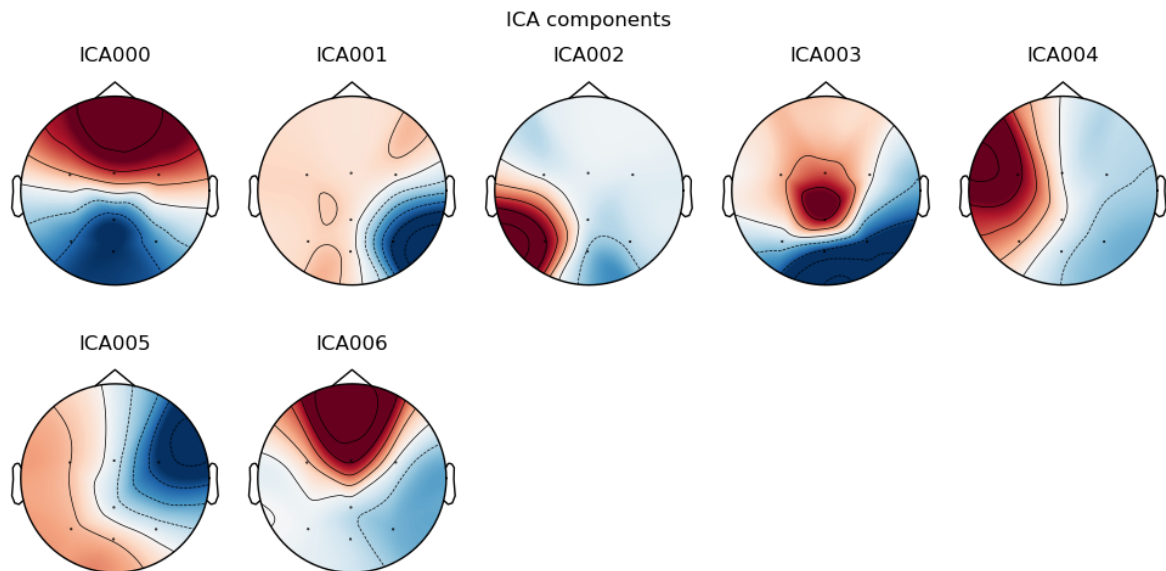
Selecting by number: 7 components

Fitting ICA took 0.0s.

****No ICA components to remove found****

```
C:\Users\IamAh\Desktop\UNIVERSITY - CSAI\Cognitive Neuroscience\M001_Cognitive_Neuroscience_Practical_Material\EEG_practical\ica_function.py:45: RuntimeWarning: This filename (C:\Users\IamAh\Desktop\UNIVERSITY - CSAI\Cognitive Neuroscience\M001_Cognitive_Neuroscience_Practical_Material\EEG_practical\3. ICA\Open_eyes_2_ica.fif) does not conform to MNE naming conventions. All raw files should end with raw.fif, raw_sss.fif, raw_tsss.fif, _meg.fif, _eeg.fif, _ieeg.fif, raw.fif.gz, raw_sss.fif.gz, raw_tsss.fif.gz, _meg.fif.gz, _eeg.fif.gz or _ieeg.fif.gz
```

```
raw.save(fname = "3. ICA/"+new_name, overwrite = True)
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 2 out of 2 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 3 out of 3 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 4 out of 4 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 8 out of 8 | elapsed: 0.0s finished
```



Applying ICA to Raw instance

Transforming to ICA space (7 components)

Zeroing out 0 ICA components

Projecting back using 7 PCA components

```
Writing C:\Users\IamAh\Desktop\UNIVERSITY - CSAI\Cognitive Neuroscience\M001_Cognitive_Neuroscience_Practical_Material\EEG_practical\3. ICA\vision.csv__ica.fif
```

```
Closing C:\Users\IamAh\Desktop\UNIVERSITY - CSAI\Cognitive Neuroscience\M001_Cognitive_Neuroscience_Practical_Material\EEG_practical\3. ICA\vision.csv__ica.fif
```

```
[done]
```

```
C:\Users\IamAh\Desktop\UNIVERSITY - CSAI\Cognitive Neuroscience\M001_Cognitive_Neuroscience_Practical_Material\EEG_practical\ica_function.py:45: RuntimeWarning: This filename (C:\Users\IamAh\Desktop\UNIVERSITY - CSAI\Cognitive Neuroscience\M001_Cognitive_Neuroscience_Practical_Material\EEG_practical\3. ICA\vision.csv__ica.fif) does not conform to MNE naming conventions. All raw files should end with raw.fif, raw_sss.fif, raw_tsss.fif, _meg.fif, _eeg.fif, _ieeg.fif, raw.fif.gz, raw_sss.fif.gz, raw_tsss.fif.gz, _meg.fif.gz, _eeg.fif.gz or _ieeg.fif.gz
```

```
raw.save(fname = "3. ICA/"+new_name, overwrite = True)
```

Step 4: Adding annotations manually

Up until now, the raw data does not contain any markers, meaning that each file is a continuous flow of EEG data without any marker indicating where the open_eyes and closed_eyes trials happen.

That is normal, since we recorded the two conditions over the whole file, not at specific times during the recording.

Nevertheless, it is better to split the whole EEG recording into smaller parts. For example, since each recording lasts 60 seconds, we could split it into 12 sub-parts of 5 seconds each, and tell the code that each of those 12 sub-parts marks the beginning of a 5-seconds long open/closed_eyes trial, depending on the file.

This will allow us to "Epoch" the data, which will make our life easier later in the notebook!

In [12]:

```
os.getcwd()
```

Out[12]:

```
'C:\\Users\\IamAh\\Desktop\\UNIVERSITY - CSAI\\Cognitive Neuroscience\\M001_Cognitive_Neuroscience_Practical_Material\\EEG_practical'
```

In [13]:

```
files_open_eyes = os.listdir("3. ICA") # path to where the ICA files are stored
```

```
In [14]: for file_open in files_open_eyes:
    if "Open" in file_open:
        sample_data_raw_file = "3. ICA/"+file_open
        raw_open_eyes = mne.io.read_raw_fif(sample_data_raw_file,
preload= True)

        # Annotation

        my_annot_open = mne.Annotations(onset=[0, 5, 10, 15, 20, 25, 30,
35, 40, 45, 50, 55], # in seconds
duration=[5,5,5,5,5,5,5,5,5,5,5,5], # in
seconds, too
description=['eye_open', 'eye_open',
'eye_open', 'eye_open', 'eye_open', 'eye_open', 'eye_open',
'eye_open', 'eye_open', 'eye_open', 'eye_open' ])
        print(my_annot_open)

        # Adding markers

        raw_open_eyes.set_annotations(my_annot_open)
        print(raw_open_eyes.annotations)

        # convert meas_date (a tuple of seconds, microseconds) into a
float:
        meas_date = raw_open_eyes.info['meas_date']
        orig_time = raw_open_eyes.annotations.orig_time
        print(meas_date == orig_time)

        fig = raw_open_eyes.plot_psd(fmax = 40)

        raw_open_eyes.save("4. Epoch/
Annotations/"+file_open[0:11]+"_annot_raw.fif", overwrite = True)
```

```
Opening raw data file 3. ICA/Open_eyes_1_ica.fif...
  Range : 0 ... 16455 =      0.000 ...    65.820 secs
Ready.
Reading 0 ... 16455 =      0.000 ...    65.820 secs...
<Annotations | 12 segments: eye_open (12)>
<Annotations | 12 segments: eye_open (12)>
True
NOTE: plot_psd() is a legacy function. New code should use .compute_psd().plot().
Effective window size : 8.192 (s)
Writing C:\Users\IamAh\Desktop\UNIVERSITY - CSAI\Cognitive Neuroscience\M001_Cognitive_Neuroscience_Practical_Material\EEG_practical\4. Epoch\Annotations\Open_eyes_1_annot_raw.fif

C:\Users\IamAh\AppData\Local\Temp\ipykernel_21112\2234308319.py:4: RuntimeWarning:
This filename (3. ICA/Open_eyes_1_ica.fif) does not conform to MNE naming conventions. All raw files should end with raw.fif, raw_sss.fif, raw_tsss.fif, _meg.fif, _eeg.fif, _ieeg.fif, raw.fif.gz, raw_sss.fif.gz, raw_tsss.fif.gz, _meg.fif.gz, _eeg.fif.gz or _ieeg.fif.gz
  raw_open_eyes = mne.io.read_raw_fif(sample_data_raw_file, preload= True)
C:\Users\IamAh\anaconda3\Lib\site-packages\mne\viz\utils.py:165: UserWarning: Matplotlib is currently using module://matplotlib_inline.backend_inline, which is a non-GUI backend, so cannot show the figure.
  (fig or plt).show(**kwargs)
Closing C:\Users\IamAh\Desktop\UNIVERSITY - CSAI\Cognitive Neuroscience\M001_Cognitive_Neuroscience_Practical_Material\EEG_practical\4. Epoch\Annotations\Open_eyes_1_annot_raw.fif
[done]
Opening raw data file 3. ICA/Open_eyes_2_ica.fif...
  Range : 0 ... 16295 =      0.000 ...    65.180 secs
Ready.
Reading 0 ... 16295 =      0.000 ...    65.180 secs...
<Annotations | 12 segments: eye_open (12)>
<Annotations | 12 segments: eye_open (12)>
True
NOTE: plot_psd() is a legacy function. New code should use .compute_psd().plot().
Effective window size : 8.192 (s)
C:\Users\IamAh\AppData\Local\Temp\ipykernel_21112\2234308319.py:4: RuntimeWarning:
This filename (3. ICA/Open_eyes_2_ica.fif) does not conform to MNE naming conventions. All raw files should end with raw.fif, raw_sss.fif, raw_tsss.fif, _meg.fif, _eeg.fif, _ieeg.fif, raw.fif.gz, raw_sss.fif.gz, raw_tsss.fif.gz, _meg.fif.gz, _eeg.fif.gz or _ieeg.fif.gz
  raw_open_eyes = mne.io.read_raw_fif(sample_data_raw_file, preload= True)
Writing C:\Users\IamAh\Desktop\UNIVERSITY - CSAI\Cognitive Neuroscience\M001_Cognitive_Neuroscience_Practical_Material\EEG_practical\4. Epoch\Annotations\Open_eyes_2_annot_raw.fif
Closing C:\Users\IamAh\Desktop\UNIVERSITY - CSAI\Cognitive Neuroscience\M001_Cognitive_Neuroscience_Practical_Material\EEG_practical\4. Epoch\Annotations\Open_eyes_2_annot_raw.fif
[done]
```

```
C:\Users\IamAh\anaconda3\Lib\site-packages\mne\viz\utils.py:165: UserWarning: Matplotlib is currently using module://matplotlib_inline.backend_inline, which is a non-GUI backend, so cannot show the figure.
```

```
(fig or plt).show(**kwargs)
```

```
In [15]: files_closed_eyes = os.listdir("3. ICA") # path to where the ICA files  
are stored
```

```
In [16]: for file_closed in files_closed_eyes:
    if "Closed" in file_closed:
        sample_data_raw_file = "3. ICA/"+file_closed
        raw_closed_eyes = mne.io.read_raw_fif(sample_data_raw_file,
preload= True)

        # Annotation

        my_annot_closed = mne.Annotations(onset=[0, 5, 10, 15, 20, 25,
30, 35, 40, 45, 50, 55], # in seconds
duration=[5,5,5,5,5,5,5,5,5,5,5,5], # in
seconds, too
description=['eye_closed', 'eye_closed',
'eye_closed','eye_closed','eye_closed', 'eye_closed','eye_closed',
'eye_closed', 'eye_closed','eye_closed','eye_closed','eye_closed' ])
        print(my_annot_closed)

        # Adding markers

        raw_closed_eyes.set_annotations(my_annot_closed)
        print(raw_closed_eyes.annotations)

        # convert meas_date (a tuple of seconds, microseconds) into a
float:
        meas_date = raw_closed_eyes.info['meas_date']
        orig_time = raw_closed_eyes.annotations.orig_time
        print(meas_date == orig_time)

        fig = raw_closed_eyes.plot_psd(fmax = 40)

        raw_closed_eyes.save("4. Epoch/
Annotations/"+file_closed[0:13]+"_annot_raw.fif", overwrite = True)
```

```
Opening raw data file 3. ICA/Closed_eyes_1_ica.fif...
  Range : 0 ... 16279 =      0.000 ...    65.116 secs
Ready.
Reading 0 ... 16279 =      0.000 ...    65.116 secs...
<Annotations | 12 segments: eye_closed (12)>
<Annotations | 12 segments: eye_closed (12)>
True
NOTE: plot_psd() is a legacy function. New code should use .compute_psd().plot().
Effective window size : 8.192 (s)

C:\Users\IamAh\AppData\Local\Temp\ipykernel_21112\2187976207.py:4: RuntimeWarning:
This filename (3. ICA/Closed_eyes_1_ica.fif) does not conform to MNE naming convent
ions. All raw files should end with raw.fif, raw_sss.fif, raw_tsss.fif, _meg.fif, _
eeg.fif, _ieeg.fif, raw.fif.gz, raw_sss.fif.gz, raw_tsss.fif.gz, _meg.fif.gz, _eeg.
fif.gz or _ieeg.fif.gz
  raw_closed_eyes = mne.io.read_raw_fif(sample_data_raw_file, preload= True)
C:\Users\IamAh\anaconda3\Lib\site-packages\mne\viz\utils.py:165: UserWarning: Matpl
otlib is currently using module://matplotlib_inline.backend_inline, which is a non-
GUI backend, so cannot show the figure.
  (fig or plt).show(**kwargs)
Writing C:\Users\IamAh\Desktop\UNIVERSITY - CSAI\Cognitive Neuroscience\M001_Cognit
ive_Neuroscience_Practical_Material\EEG_practical\4. Epoch\Annotations\Closed_eyes_
1_annot_raw.fif
Closing C:\Users\IamAh\Desktop\UNIVERSITY - CSAI\Cognitive Neuroscience\M001_Cognit
ive_Neuroscience_Practical_Material\EEG_practical\4. Epoch\Annotations\Closed_eyes_
1_annot_raw.fif
[done]
Opening raw data file 3. ICA/Closed_eyes_2_ica.fif...
  Range : 0 ... 16303 =      0.000 ...    65.212 secs
Ready.
Reading 0 ... 16303 =      0.000 ...    65.212 secs...
<Annotations | 12 segments: eye_closed (12)>
<Annotations | 12 segments: eye_closed (12)>
True
NOTE: plot_psd() is a legacy function. New code should use .compute_psd().plot().
C:\Users\IamAh\AppData\Local\Temp\ipykernel_21112\2187976207.py:4: RuntimeWarning:
This filename (3. ICA/Closed_eyes_2_ica.fif) does not conform to MNE naming convent
ions. All raw files should end with raw.fif, raw_sss.fif, raw_tsss.fif, _meg.fif, _
eeg.fif, _ieeg.fif, raw.fif.gz, raw_sss.fif.gz, raw_tsss.fif.gz, _meg.fif.gz, _eeg.
fif.gz or _ieeg.fif.gz
  raw_closed_eyes = mne.io.read_raw_fif(sample_data_raw_file, preload= True)
Effective window size : 8.192 (s)
Writing C:\Users\IamAh\Desktop\UNIVERSITY - CSAI\Cognitive Neuroscience\M001_Cognit
ive_Neuroscience_Practical_Material\EEG_practical\4. Epoch\Annotations\Closed_eyes_
2_annot_raw.fif
Closing C:\Users\IamAh\Desktop\UNIVERSITY - CSAI\Cognitive Neuroscience\M001_Cognit
ive_Neuroscience_Practical_Material\EEG_practical\4. Epoch\Annotations\Closed_eyes_
2_annot_raw.fif
[done]
```

```
C:\Users\IamAh\anaconda3\Lib\site-packages\mne\viz\utils.py:165: UserWarning: Matplotlib is currently using module://matplotlib_inline.backend_inline, which is a non-GUI backend, so cannot show the figure.
```

```
(fig or plt).show(**kwargs)
```

Step 5: epoching

```
In [17]: from Epoching import epoching
```

First of all, you're doing great! Congratulations for getting this far, we are almost there!

In the next step, epoching, you will "epoch" your files: this means that the program will take each file and extract the 5-seconds long trials marked by the markers we inserted in the Annotations step and save this information in a new, "Epoch" file, which can be thought of as a N-dimensional numpy array of shape: trials(the amount of markers we inserted in the Annotations step) X nr_channels X time points.

```
In [18]: os.getcwd()
```

```
Out[18]: 'C:\\Users\\IamAh\\Desktop\\UNIVERSITY - CSAI\\Cognitive Neuroscience\\M001_Cognitive Neuroscience Practical Material\\EEG_practical'
```

```
In [22]: for file in os.listdir("4. Epoch/Annotations"):
          if "raw" in file:
              epoching(file, "4. Epoch/Annotations")
```

```
Opening raw data file 4. Epoch/Annotations/Closed_Eyes_1_annot_raw.fif...
  Range : 0 ... 16279 =      0.000 ...    65.116 secs
Ready.
Not setting metadata
12 matching events found
Setting baseline interval to [-1.0, 0.0] s
Applying baseline correction (mode: mean)
0 projection items activated
Loading data for 12 events and 1501 original time points ...
1 bad epochs dropped
Opening raw data file 4. Epoch/Annotations/Closed_Eyes_2_annot_raw.fif...
  Range : 0 ... 16303 =      0.000 ...    65.212 secs
Ready.
Not setting metadata
12 matching events found
Setting baseline interval to [-1.0, 0.0] s
Applying baseline correction (mode: mean)
0 projection items activated
Loading data for 12 events and 1501 original time points ...
1 bad epochs dropped
Opening raw data file 4. Epoch/Annotations/Open_Eyes_1_annot_raw.fif...
  Range : 0 ... 16455 =      0.000 ...    65.820 secs
Ready.
Not setting metadata
12 matching events found
Setting baseline interval to [-1.0, 0.0] s
Applying baseline correction (mode: mean)
0 projection items activated
Loading data for 12 events and 1501 original time points ...
1 bad epochs dropped
Opening raw data file 4. Epoch/Annotations/Open_Eyes_2_annot_raw.fif...
  Range : 0 ... 16295 =      0.000 ...    65.180 secs
Ready.
Not setting metadata
12 matching events found
Setting baseline interval to [-1.0, 0.0] s
Applying baseline correction (mode: mean)
0 projection items activated
Loading data for 12 events and 1501 original time points ...
1 bad epochs dropped
```

Load two of the files you epoched to make sure the time ranges are correct: the files should report: $t = -1000.00 \dots 5000\text{ms}$ >> from -1 to 5 seconds.

```
In [23]: epoch_E0 = mne.read_epochs("4. Epoch/Open/Open_Eyes_2-  
open_eyes_epo.fif", preload=True)
```

```
Reading C:\Users\IamAh\Desktop\UNIVERSITY - CSAI\Cognitive Neuroscience\M001_Cognit  
ive_Neuroscience_Practical_Material\EEG_practical\4. Epoch\Open\Open_Eyes_2-open_ey  
es_epo.fif ...
```

```
Found the data of interest:
```

```
t = -1000.00 ... 5000.00 ms
```

```
0 CTF compensation matrices available
```

```
Not setting metadata
```

```
11 matching events found
```

```
No baseline correction applied
```

```
0 projection items activated
```

Check the shape of the epoch by converting it into an array:

```
In [24]: epoch_E0.get_data().shape
```

```
C:\Users\IamAh\AppData\Local\Temp\ipykernel_21112\2105429179.py:1: FutureWarning: T  
he current default of copy=False will change to copy=True in 1.7. Set the value of  
copy explicitly to avoid this warning
```

```
epoch_E0.get_data().shape
```

```
Out[24]: (11, 8, 1501)
```

We have 8 channels and 1501 time points in the epoch. The first dimension (the number of trial markers) could be either 12 (11 in the epoch, since index starts at 0 in array) or slightly less, since during the epoching process, the function might have dropped a few "bad epochs".

```
In [25]: epoch_EC = mne.read_epochs("4. Epoch/Closed/Closed_Eyes_2-  
closed_eyes_epo.fif", preload=True)
```

```
Reading C:\Users\IamAh\Desktop\UNIVERSITY - CSAI\Cognitive Neuroscience\M001_Cognit  
ive_Neuroscience_Practical_Material\EEG_practical\4. Epoch\Closed\Closed_Eyes_2-clo  
sed_eyes_epo.fif ...
```

```
Found the data of interest:
```

```
t = -1000.00 ... 5000.00 ms
```

```
0 CTF compensation matrices available
```

```
Not setting metadata
```

```
11 matching events found
```

```
No baseline correction applied
```

```
0 projection items activated
```

Same for this epoch:

```
In [26]: epoch_EC.get_data().shape
```

```
C:\Users\IamAh\AppData\Local\Temp\ipykernel_21112\3817631679.py:1: FutureWarning: The current default of copy=False will change to copy=True in 1.7. Set the value of copy explicitly to avoid this warning
```

```
epoch_EC.get_data().shape
```

```
Out[26]: (11, 8, 1501)
```

Step 6: power spectra

Let's visualise the power spectra of the two loaded epoched files, just to see if there is any difference between the two conditions. Can you see any difference between the two plots? Does one condition report larger/smaller values than the other and if so, in what frequency range? The dotted vertical lines that you see in the plots represent the frequency range (1-30 Hz) that we are interested in.

```
In [45]: print(epoch_E0.__dir__())
```

```
['_allow_empty', 'event_id', 'selection', 'events', '_metadata', 'drop_log', 'detrend', '_raw', 'picks', 'info', '_current', 'preload', '_data', '_do_baseline', '_offset', '_raw_times', '_times_readonly', 'reject_tmin', 'reject_tmax', '_decim', '_decim_slice', 'baseline', 'reject', 'flat', '_bad_dropped', '_channel_type_idx', '_reject_time', '_do_delayed_proj', '_projector', '_filename', '_raw_sfreq', '_annotations', '_unsafe_annot_add', '__module__', '__doc__', '__init__', '_get_epoch_from_raw', '_check_consistency', 'reset_drop_log_selection', 'load_data', 'apply_baseline', '_reject_setup', '_is_good_epoch', '_detrend_offset_decim', 'iter_evoked', 'subtract_evoked', 'average', 'standard_error', '_compute_aggregate', '_name', '_get_name', '_evoked_from_epoch_data', 'ch_names', 'plot', 'plot_topo_image', 'drop_bad', 'drop_log_stats', 'plot_drop_log', 'plot_image', 'drop', '_project_epoch', '_handle_empty', '_get_data', '_data_sel_copy_scale', '_detrend_picks', 'get_data', 'apply_function', 'filename', '__repr__', '_repr_html_', 'crop', 'copy', '__deepcopy__', 'save', 'export', 'equalize_event_counts', 'compute_psd', 'plot_psd', 'to_data_frame', 'as_type', 'proj', 'add_proj', 'apply_proj', 'del_proj', 'plot_projs_topomap', '_reconstruct_proj', '__dict__', '__weakref__', '__new__', '__hash__', '__str__', '__getattr__', '__setattr__', '__delattr__', '__lt__', '__le__', '__eq__', '__ne__', '__gt__', '__ge__', '__reduce_ex__', '__reduce__', '__getstate__', '__subclasshook__', '__init_subclass__', '__format__', '__sizeof__', '__dir__', '__class__', '__contains__', 'compensation_grade', 'get_channel_types', 'pick_types', 'pick_channels', 'pick', 'reorder_channels', 'drop_channels', '_pick_drop_channels', '_pick_projs', 'add_channels', 'add_reference_channels', 'set_eeg_reference', 'get_montage', 'set_montage', '_get_channel_positions', '_set_channel_positions', 'set_channel_types', 'rename_channels', 'plot_sensors', 'anonymize', 'set_meas_date', 'interpolate_bads', 'savgol_filter', 'filter', 'resample', 'apply_hilbert', 'tmin', 'tmax', 'decimate', 'shift_time', '_update_first_last', 'time_as_index', '_handle_tmin_tmax', 'times', '_set_times', '_size', '_getitem_', '_item_to_select', '_getitem', '_keys_to_idx', '_len_', '_iter_', '_next_', '_stop_iter', 'next', '_check_metadata', 'metadata', 'annotations', 'set_annotations', 'get_annotations_per_epoch', 'add_annotations_to_metadata', 'plot_psd_topo', 'plot_psd_topomap', '_set_legacy_nfft_default']
```

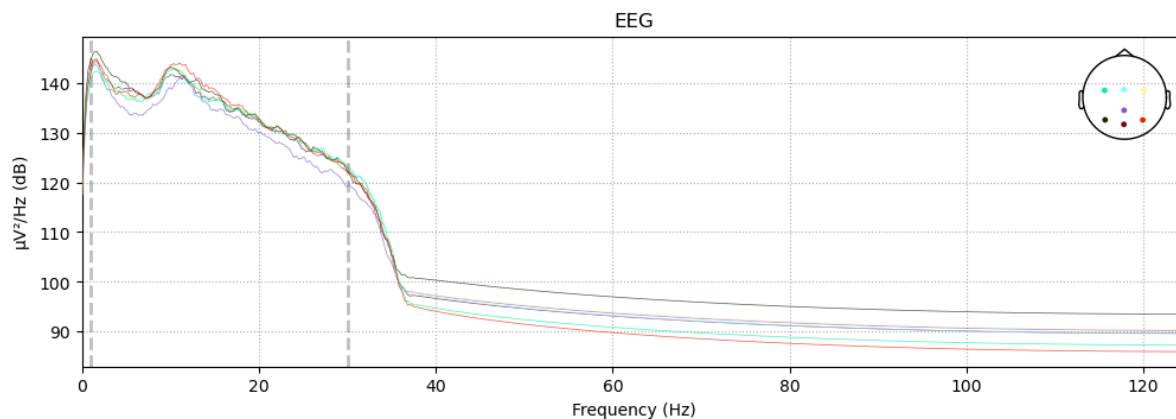
In [27]: `epoch_E0.compute_psd().plot(picks='eeg')`

Using multitaper spectrum estimation with 7 DPSS windows
Averaging across epochs...

C:\Users\IamAh\anaconda3\Lib\site-packages\mne\viz\utils.py:165: UserWarning: Matplotlib is currently using module://matplotlib_inline.backend_inline, which is a non-GUI backend, so cannot show the figure.

(fig or plt).show(**kwargs)

Out[27]:



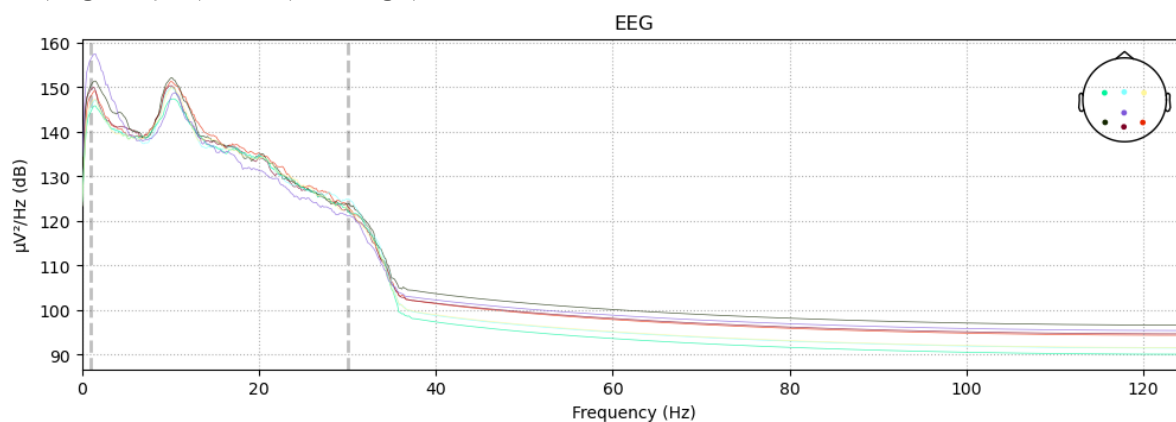
In [28]: `epoch_EC.compute_psd().plot(picks='eeg')`

Using multitaper spectrum estimation with 7 DPSS windows
Averaging across epochs...

C:\Users\IamAh\anaconda3\Lib\site-packages\mne\viz\utils.py:165: UserWarning: Matplotlib is currently using module://matplotlib_inline.backend_inline, which is a non-GUI backend, so cannot show the figure.

(fig or plt).show(**kwargs)

Out[28]:



Step 7: TFR visualization

In [29]: `from TFR import tfr_analysis
from matplotlib import colors
import matplotlib.pyplot as plt`

And now, the last step: time-frequency visualization.

In this last step, we will work with the two epoched groups of files, open vs closed.

What we want to do is, extracting and visualising the "*temporal evolution of spectral properties of the signal*" (https://link.springer.com/referenceworkentry/10.1007/978-1-4614-6675-8_421).

In order to do so, you should get familiar with the concept of the "**Fourier Transform**": this is a process that, by fitting sinusoidal waves of different frequencies to your cleaned and epoched EEG signal, it extracts the "similarity" of the signal to these different sine waves, transposing the signal to a time-frequency representation. The signal can then be visualised on the time domain (the time window you set during the epoching) and a frequency range (what frequency range are we interested in? In this case, alpha range > 8-12).

This means that, for the epoched files, we would like to explore (in our case) the alpha properties for the corresponding signals at certain channels, over time.

Alpha waves are known to inhibit areas of the cortex not in use (i.e., occipital areas when eyes are closed) (first observed by Berger, H. in 1929), therefore, the expectation is that, in the "closed eyes" condition, alpha waves will be stronger (inhibiting the occipital lobe) compared to the "open eyes" condition.

Access the link below to see an animation of the Fourier Transform:

Source for animation: https://commons.wikimedia.org/wiki/File:Fourier_transform_time_and_frequency_domains.gif

```
In [30]: channels = ["Fz", "C3", "Cz", "C4", "Pz", "P07", "Oz", "P08"]
```

```
In [31]: min_list = []
max_list = []

sd_min_list = []
sd_max_list = []
```

In [32]:

```
freq_ranges = ["Alpha"]

times = np.linspace(start = 0, stop = 5, num = 151) # tmax equals 5 seconds

for iChan in range(len(channels)):

    info_list_fr = [4] # range between fmin and fmax
    info_list_fmin = [8] # fmin Alpha
    info_list_fmax = [12] # fmax Alpha

    for iBand in range(len(freq_ranges)): # Len = 1

        freq_range, fmin, fmax = info_list_fr[iBand],
info_list_fmin[iBand], info_list_fmax[iBand]
        freqs = np.arange(fmin, fmax, 1.)

        # Participant 1

        files = tfr_analysis(["Open_Eyes_1-open_eyes_epo.fif",
"Open_Eyes_2-open_eyes_epo.fif" ], ["Closed_Eyes_1-closed_eyes_epo.fif",
"Closed_Eyes_2-closed_eyes_epo.fif"], "4. Epoch/Open", "4. Epoch/
Closed", 2, freq_range, fmin, fmax, channels[iChan]) # call to funciton tfr_analysis

        final_to_plot_open = files[0]
        final_to_plot_closed = files[1]

        p_open = final_to_plot_open[iChan]
        p_closed = final_to_plot_closed[iChan]

        # Take the Log

        p_open = np.log(p_open)
        p_closed = np.log(p_closed)
```

```
# Extract min/max values for colorbar ranges:
```

```
max_list.append(np.amax(p_open))
max_list.append(np.amax(p_closed))
min_list.append(np.amin(p_open))
min_list.append(np.amin(p_closed))

sd_max_list.append(np.amax(p_open-p_closed))
sd_min_list.append(np.amin(p_open-p_closed))
```

```
# Final Lists
```

```
p_open_list = [p_open]
p_closed_list = [p_closed]
p_spec_diff_list = [p_open-p_closed]
```

```
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
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[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
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[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
```

[illegible]

```
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished
```

Sort the lists:

```
In [33]: min_list.sort()
max_list.sort()

sd_min_list.sort()
sd_max_list.sort()
```

From each of them, extract the 25th (for the min value) and the 75th (for the max value) percentiles:

```
In [34]: vmin= np.percentile(min_list, 25)
vmax= np.percentile(max_list, 75)

vmin_sd= np.percentile(sd_min_list, 25)
vmax_sd= np.percentile(sd_max_list, 75)
```

Run the code again, this time with vmin, vmax etc... already pre-set:

```
In [35]: freq_ranges = ["Alpha"]

times = np.linspace(start = 0, stop = 5, num = 151) # tmax equals 5
seconds

# PLOTTING:

for iChan in range(len(channels)):

    print("Visualising channel {}".format(channels[iChan]))

    info_list_fr = [4] # range between fmin and fmax
    info_list_fmin = [8] # fmin Alpha
    info_list_fmax = [12] # fmax Alpha

    for iBand in range(len(freq_ranges)): # Len = 1

        freq_range, fmin, fmax = info_list_fr[iBand],
info_list_fmin[iBand], info_list_fmax[iBand]
        freqs = np.arange(fmin, fmax, 1.)

        # Participant 1

        files = tfr_analysis(["Open_Eyes_1-open_eyes_epo.fif",
"Open_Eyes_2-open_eyes_epo.fif" ], ["Closed_Eyes_1-closed_eyes_epo.fif",
"Closed_Eyes_2-closed_eyes_epo.fif"],
                             "4. Epoch/Open", "4. Epoch/Closed", 2,
freq_range, fmin, fmax, channels[iChan]) # call to funciton tfr_analysis

        final_to_plot_open = files[0]
        final_to_plot_closed = files[1]

        p_open = final_to_plot_open[iChan]
        p_closed = final_to_plot_closed[iChan]
```

```
# Take the Log

p_open = np.log(p_open)
p_closed = np.log(p_closed)

# Final lists
p_open_list = [p_open]
p_closed_list = [p_closed]
p_spec_diff_list = [p_open-p_closed]

for j in range(len(p_open_list)):

    fig, axes = plt.subplots(nrows=1, ncols=3, figsize = (20,4))

    plt.axes(axes[0])
    plt.imshow(p_open_list[j], extent = [times[0], times[-1],
freqs[0], freqs[-1]], cmap = "RdBu", aspect='auto', origin='lower', vmin
= vmin, vmax = vmax)
    plt.title("Open eyes")
    plt.colorbar()

    plt.axes(axes[1])
    plt.imshow(p_closed_list[j], extent = [times[0], times[-1],
freqs[0], freqs[-1]], cmap = "RdBu", aspect='auto',origin='lower', vmin
= vmin, vmax = vmax)
    plt.title("Closed eyes")
    plt.colorbar()

    plt.axes(axes[2])
    plt.imshow(p_spec_diff_list[j], extent = [times[0],
times[-1], freqs[0], freqs[-1]],cmap = "RdBu",
aspect='auto',origin='lower', vmin = vmin_sd, vmax = vmax_sd)
    plt.title("Spectral difference between two conditions")
    plt.colorbar()
```

```

fig.subplots_adjust(right=0.8)

plt.show()

# Saving files to numpy arrays
np.save("5. TFR/TFR_open_alpha_"+channels[iChan]
,p_open_list[j])
np.save("5. TFR/
TFR_closed_alpha_"+channels[iChan],p_closed_list[j])

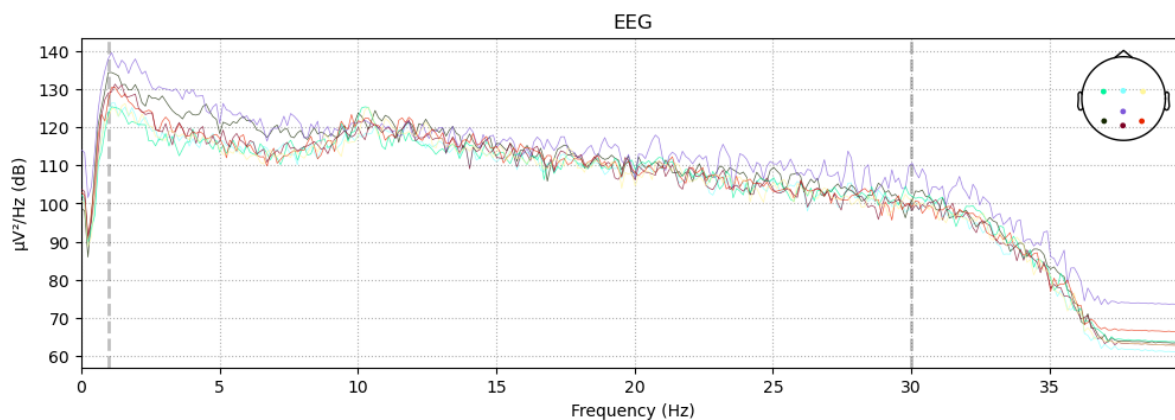
```

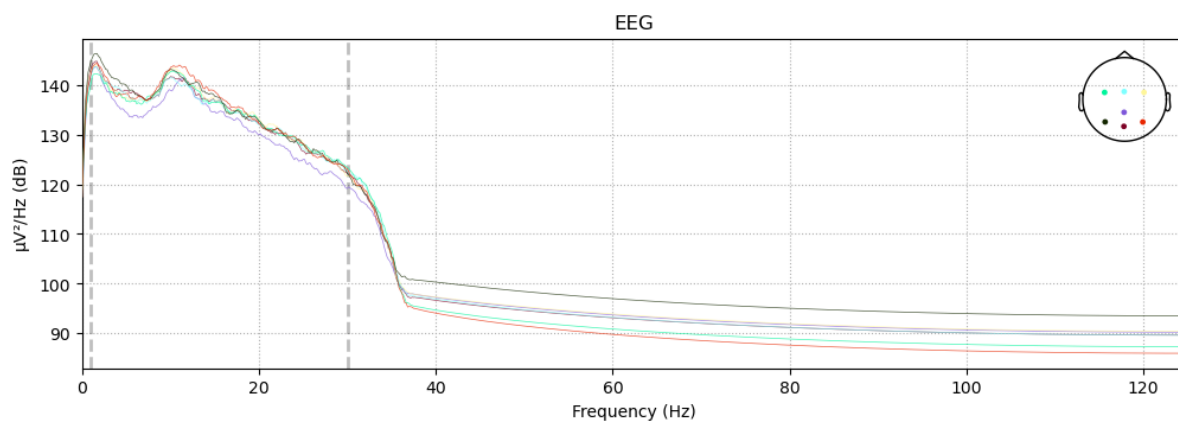
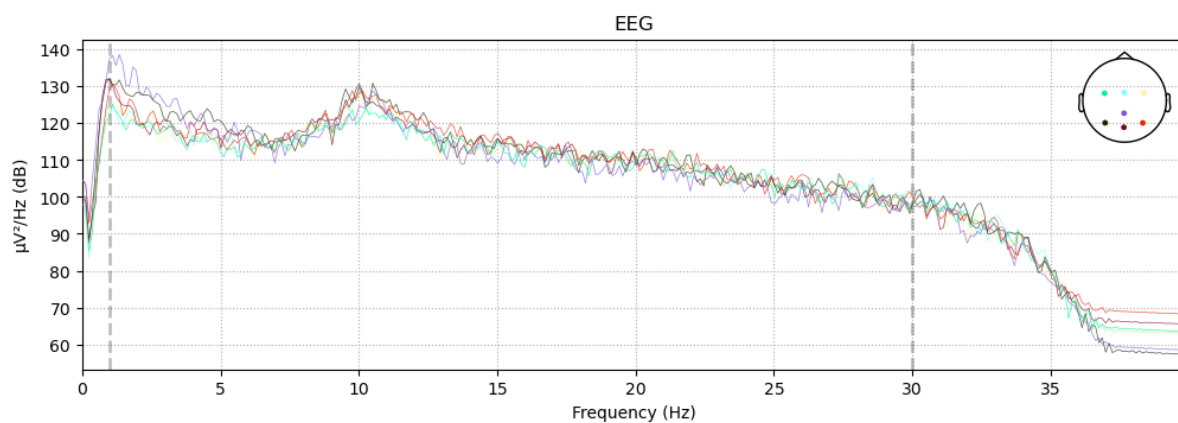
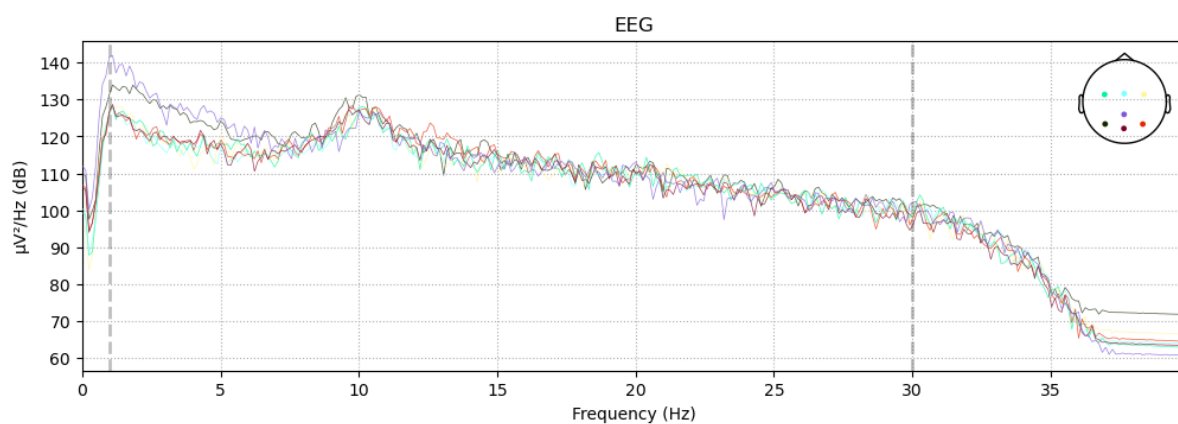
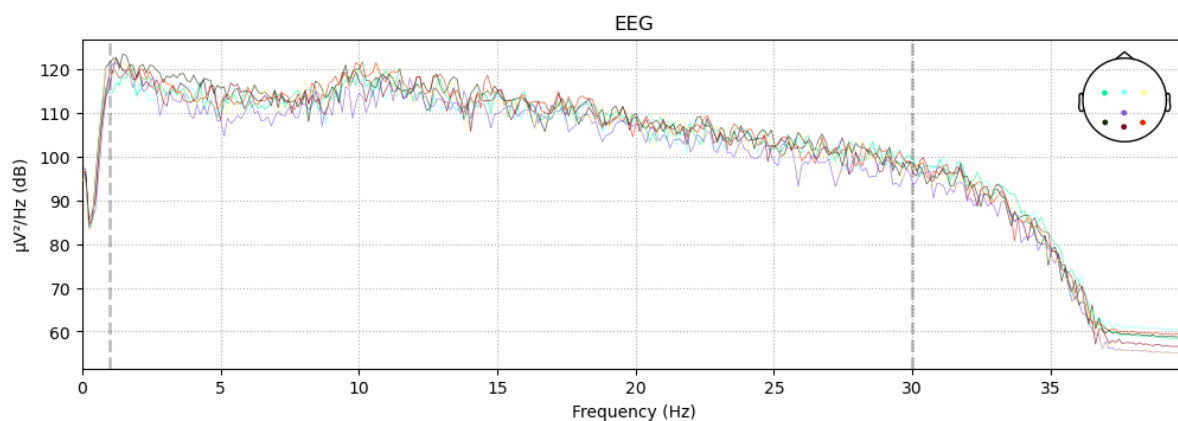
Visualising channel Fz

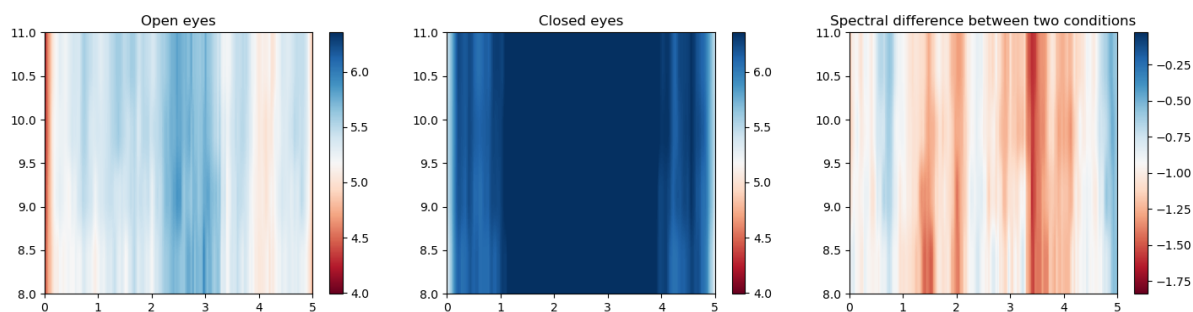
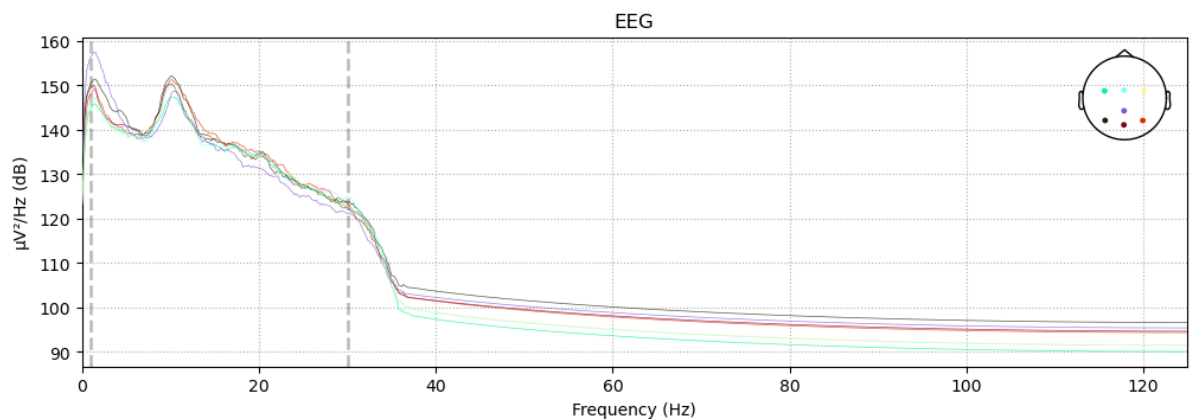
```

[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished

```

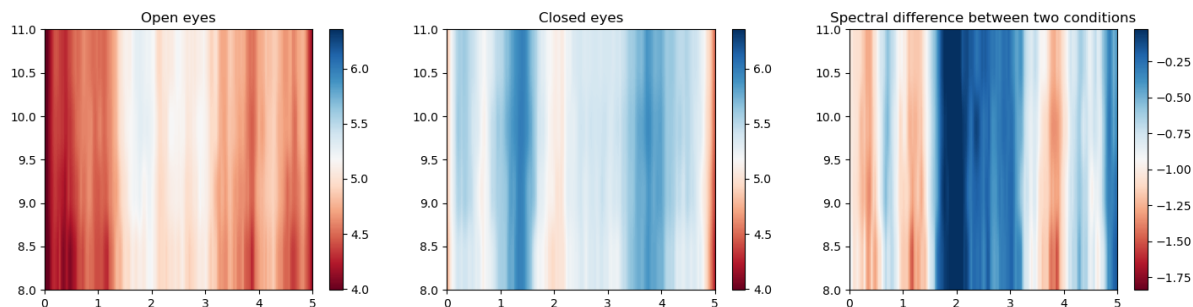






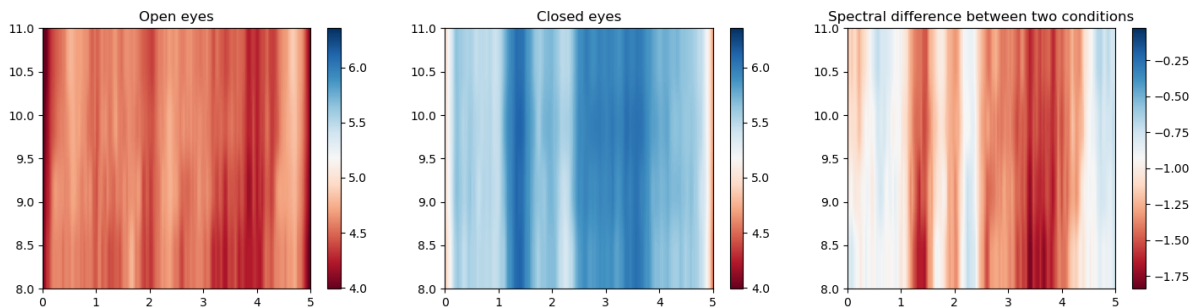
Visualising channel C3

```
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished
```



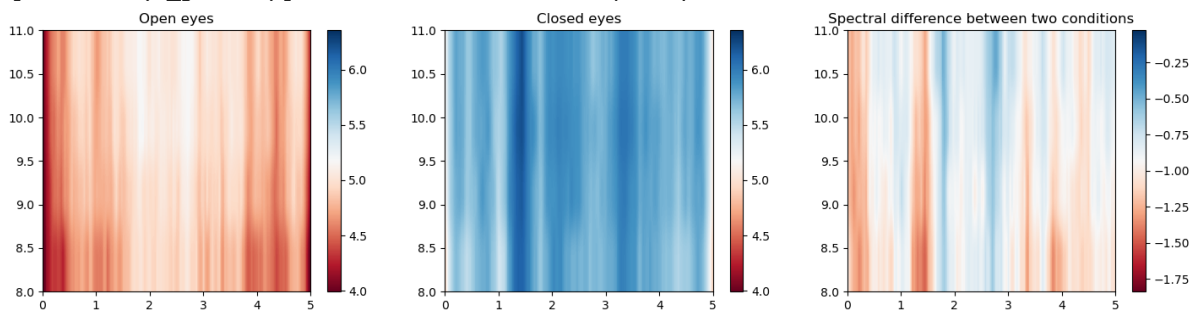
Visualising channel Cz

```
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished
```



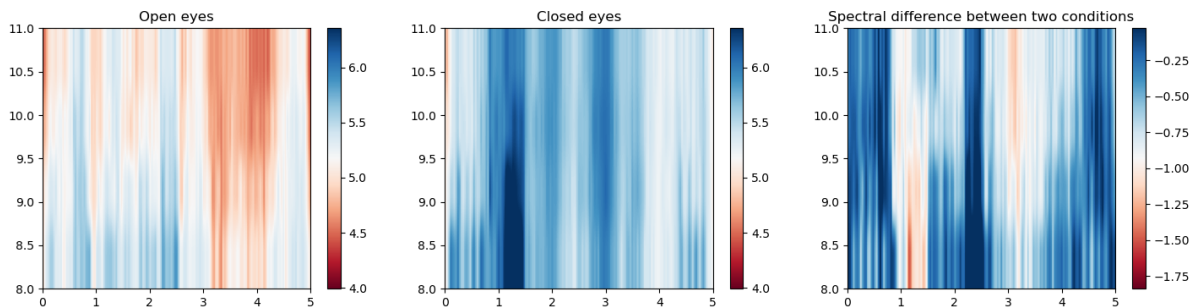
Visualising channel C4

```
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished
```



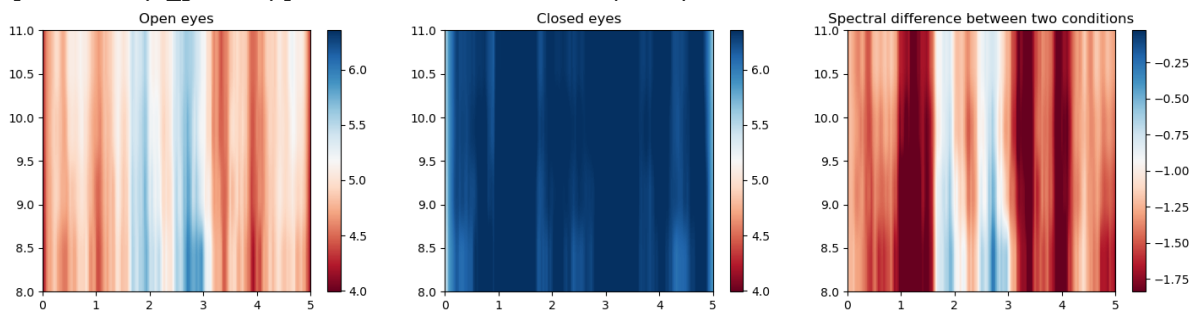
Visualising channel Pz

```
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished
```



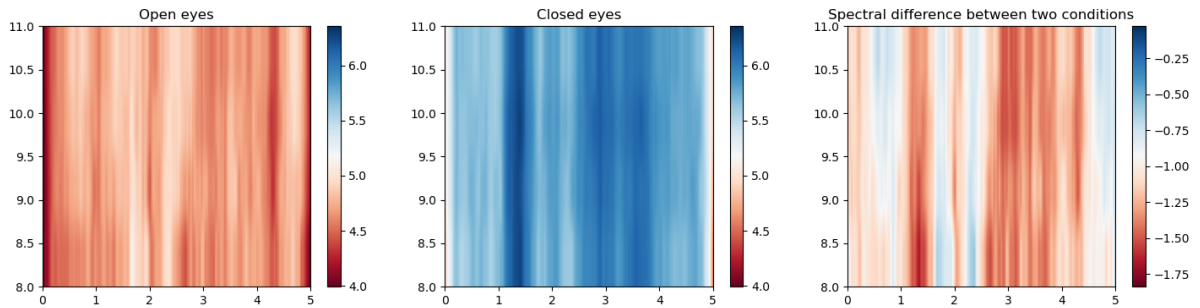
Visualising channel P07

```
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished
```



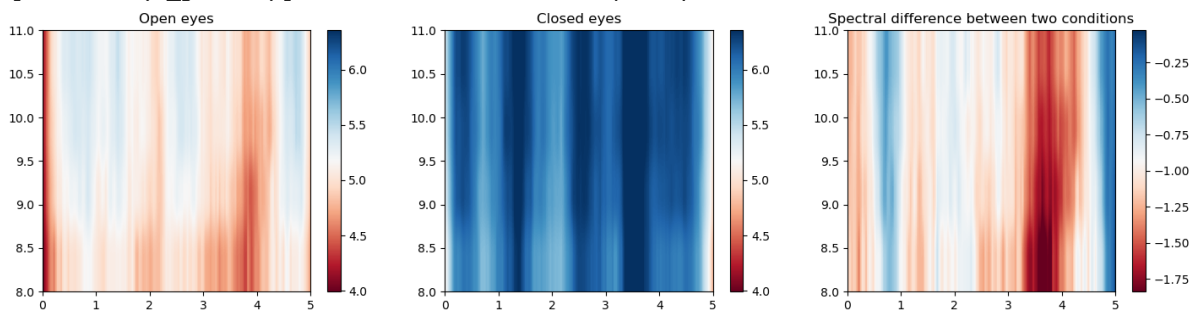
Visualising channel Oz

```
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished
```



Visualising channel P08

```
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished
[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s remaining: 0.0s
[Parallel(n_jobs=1)]: Done 1 out of 1 | elapsed: 0.0s finished
```



Step 8: Two-related samples t-test between channels-averaged conditions

```
In [36]: from scipy import stats
```

Let's load the arrays into two, grand arrays first, one for each condition and let's run a two-related samples t-test between each condition, per channel:

```
In [47]: lista = os.listdir("5. TFR")
# lista.remove('.ipynb_checkpoints')

for chan in channels:
    file_tfr_open = np.load("5. TFR/TFR_open_alpha_"+chan+".npy")
    file_tfr_closed = np.load("5. TFR/TFR_closed_alpha_"+chan+".npy")
    stat, pvalue = stats.ttest_rel(file_tfr_open, file_tfr_closed)
    print("Two-related samples t-tets between {} and {} pvalues:
    {}".format("TFR_open_alpha_"+chan, "TFR_closed_alpha_"+chan, pvalue))
```

Two-related samples t-tets between TFR_open_alpha_Fz and TFR_closed_alpha_Fz pvalue
s: [5.93400614e-04 2.33122000e-04 1.98013115e-04 8.50267725e-07
2.39566345e-05 6.09176438e-06 2.27338059e-05 9.60736796e-06
2.81945917e-05 5.27348316e-06 7.27277970e-06 5.16400738e-07
1.37386124e-05 5.26538596e-05 7.75308669e-05 2.88674918e-04
7.83969343e-04 5.18772242e-04 3.74182851e-04 5.71471663e-04
6.42889118e-04 2.44784846e-04 2.43267751e-04 3.56132094e-04
5.66957155e-04 1.06033431e-04 2.65451825e-05 2.02610462e-05
2.13496879e-04 1.05014015e-04 1.09490552e-04 2.05693660e-05
3.16006575e-06 5.29317455e-06 5.09777183e-06 3.33282144e-06
1.10171680e-05 6.30994789e-05 3.27116762e-05 1.16000999e-04
3.16392675e-04 6.10630714e-04 4.81314107e-04 8.00203023e-05
3.97273471e-05 2.01325265e-04 3.40499327e-04 5.41918859e-05
6.74836666e-05 8.45078139e-05 2.96817829e-05 6.31737790e-06
2.40775955e-05 2.04356035e-05 4.61834240e-06 5.37158304e-06
2.91909662e-06 3.09222197e-05 9.15802245e-06 9.50528615e-06
4.97818568e-05 2.00711894e-05 1.62597774e-06 1.06686345e-06
1.17093512e-05 1.61330543e-06 2.80123158e-06 1.89494458e-05
2.86173684e-05 1.60211135e-05 1.80386122e-06 1.81713262e-05
5.30863093e-05 9.42361428e-06 4.98230957e-05 2.69560706e-05
3.18360602e-05 2.72010441e-05 5.86043054e-05 8.94090168e-05
5.36585555e-04 4.09249801e-04 2.06646683e-04 2.07971964e-04
5.81872759e-05 1.26933553e-04 8.09775695e-05 3.81772965e-05
5.93864221e-05 1.01795800e-03 1.68614847e-05 3.18192449e-05
2.37132978e-04 3.20779604e-04 1.70269104e-04 3.45894899e-04
1.20734775e-03 2.11214535e-03 6.21449207e-04 8.09804278e-05
1.34845895e-04 9.23477268e-05 7.49374572e-06 7.28929598e-07
2.91790543e-05 1.29097319e-06 3.35196446e-07 2.83571098e-06
6.67534056e-06 6.06130137e-06 2.21028940e-05 1.16318575e-04
7.38469257e-06 5.18671524e-05 1.00086037e-04 8.73149673e-05
4.37982135e-06 1.11380208e-05 4.06894339e-06 1.59713785e-05
1.19384497e-05 2.94193751e-06 1.08247880e-05 1.32362037e-05
1.40281866e-04 2.26849757e-04 3.37262397e-05 1.54737508e-04
1.61049162e-05 1.78488170e-04 4.35330826e-05 1.37677518e-04
2.45672162e-05 3.60856671e-05 2.51883609e-06 4.24845740e-06
5.55362320e-06 1.48348633e-05 2.29584201e-05 5.23655555e-05
2.12063483e-04 4.60107622e-04 2.26018952e-04 2.12111983e-04
9.98247023e-04 5.02588397e-04 8.98135113e-04 2.61472675e-04
3.99973924e-05 3.56596170e-06 9.02716974e-05]

Two-related samples t-tets between TFR_open_alpha_C3 and TFR_closed_alpha_C3 pvalue
s: [1.77341114e-04 1.01127343e-05 9.53747445e-07 8.02280547e-07
2.15081573e-05 1.77474219e-05 2.56149741e-05 1.07572294e-05
1.58309316e-04 1.80534853e-06 1.56521106e-05 1.05068578e-05
3.56458752e-05 5.40817141e-05 2.57618288e-04 2.56212292e-04
2.17290241e-04 8.31780405e-04 3.51008668e-04 4.07083261e-04
3.86053885e-04 1.64208992e-03 2.53247333e-04 1.05092702e-06
2.73859225e-05 4.47835004e-05 3.15331626e-06 1.40983268e-05
7.87908507e-05 1.72276108e-04 1.77619945e-04 1.05744959e-04
5.35451550e-06 2.62143490e-05 2.64776302e-04 4.06302901e-04
2.17551033e-05 6.01342420e-06 4.08290909e-05 1.35021564e-05
3.38230644e-05 5.55829933e-05 2.20341924e-04 1.49477656e-04]

1.39058175e-05 9.79668317e-06 5.54792746e-05 1.23329723e-04
2.34730841e-04 2.82805725e-03 1.40438258e-03 3.51395507e-04
6.26055489e-03 7.58674109e-02 1.03960360e-02 1.43357646e-01
7.01769580e-01 4.19195432e-02 2.96949689e-02 3.93663705e-02
3.59401141e-02 1.21271196e-04 5.27408243e-01 1.15389647e-01
3.40884204e-03 5.37558650e-03 9.79225043e-03 1.99161642e-03
4.09219821e-03 5.20766721e-04 6.87555205e-03 1.50976073e-02
7.76983924e-03 1.31735588e-04 2.55461144e-03 3.51970441e-03
2.27369790e-03 6.12076174e-04 3.68284669e-04 6.51584924e-04
1.22132352e-03 2.28515826e-03 7.97067190e-04 1.65206139e-03
1.69345115e-04 1.83472043e-04 6.44145457e-04 4.24140587e-04
2.19898340e-05 1.43052442e-04 4.79104562e-04 3.03444152e-03
6.42873114e-04 5.36920218e-05 8.82973496e-04 1.60336283e-03
3.43758623e-03 8.33071122e-04 1.39109555e-04 2.01993925e-05
2.68962566e-05 5.42928515e-05 1.39282794e-04 1.33253158e-04
1.76127566e-04 3.11781708e-04 3.18697543e-04 7.62808709e-05
2.16441315e-04 2.11846686e-04 1.32239546e-04 1.59260699e-05
1.28187055e-04 7.42880092e-05 4.60697468e-05 1.60135395e-05
3.60162464e-04 9.73234138e-05 4.14155826e-06 1.03064810e-05
1.55814320e-07 9.01304806e-06 4.11086882e-05 9.90702908e-05
9.21870610e-06 1.03591141e-04 8.41376796e-05 6.79795631e-05
3.81002427e-04 4.13971875e-04 1.06230824e-04 1.26801577e-04
5.26975235e-05 7.15856950e-05 6.56430007e-06 3.14452720e-05
1.87930690e-04 2.40507584e-04 6.28642629e-04 3.11404882e-04
1.05624047e-04 6.00797732e-05 8.60255502e-04 9.05041098e-04
6.50466020e-04 3.43611415e-04 8.54797842e-04 4.74638259e-04
9.62913327e-03 1.32757660e-02 9.53163445e-03]

Two-related samples t-tets between TFR_open_alpha_Cz and TFR_closed_alpha_Cz pvalue

s: [1.44609032e-03 6.70058904e-04 1.46312001e-03 4.40376382e-04
2.09838037e-04 3.34825125e-04 5.82156918e-04 2.16856174e-04
1.40501033e-04 1.23352416e-04 1.49785334e-04 3.32411898e-05
7.01426933e-06 5.99420265e-05 1.72198375e-04 2.64310620e-04
1.78572936e-04 2.89260910e-04 1.32081131e-04 8.50717854e-05
9.95766142e-05 2.35556563e-04 9.95314739e-05 7.79857204e-05
1.50437010e-05 2.87286822e-05 3.54091756e-05 2.56984575e-05
3.78488265e-06 8.77119390e-05 5.17464441e-06 2.15268582e-05
1.13009677e-05 2.31824966e-06 1.72647021e-06 1.80039916e-05
1.39570707e-05 2.35907540e-05 3.07080583e-05 1.00816440e-04
1.31894822e-04 1.01620867e-04 3.81551131e-05 4.54520173e-05
2.88794554e-06 1.03552460e-05 1.71780348e-05 7.95269907e-06
7.67511606e-06 2.21260080e-04 2.58227750e-04 5.17041057e-05
2.90204814e-06 1.29628356e-05 4.88185217e-05 3.55457474e-05
3.77827615e-07 4.59068627e-06 1.46899132e-07 1.72844975e-07
6.10461712e-06 7.40386667e-06 3.94425650e-06 1.36089016e-05
6.24367834e-05 3.97231985e-06 6.19583951e-05 9.29869338e-04
2.54137615e-04 1.32441531e-04 2.97220788e-05 1.21683541e-04
3.00839966e-05 7.41380915e-07 2.19891022e-06 2.43314127e-06
6.28744914e-07 3.02504551e-05 8.59452442e-05 3.40534502e-05
1.45799422e-05 1.60268171e-05 3.00351144e-05 1.82402397e-05
1.72508625e-05 2.01992946e-06 2.37982418e-05 1.78654059e-05
2.57431326e-05 7.17722129e-06 9.55203705e-07 6.32576031e-07]

8.21459298e-06 4.61262428e-06 2.42502696e-06 8.74800982e-06
4.26788309e-06 8.92522820e-06 6.57811105e-05 1.51516312e-05
1.22640560e-06 3.27235071e-05 3.32239920e-05 3.39077101e-05
7.31649976e-06 2.32006850e-05 9.17350015e-05 2.76075160e-05
2.43684282e-06 6.33881163e-05 9.27649669e-05 4.68220996e-05
9.47807400e-06 1.98260756e-05 6.91288228e-06 2.47850141e-06
2.45694021e-06 1.53991808e-06 1.43420917e-06 8.18546833e-06
8.91348113e-06 1.03275699e-06 1.10368478e-06 7.72422696e-06
2.50549994e-05 2.86678249e-05 2.64403537e-05 8.72257138e-05
8.32620716e-05 1.59133308e-04 4.94756970e-04 4.60509645e-04
1.27327181e-04 4.19985044e-05 1.39314316e-05 9.37786457e-06
2.03762914e-05 1.24197574e-05 1.16277585e-04 2.42567281e-04
4.31231199e-04 1.87446434e-04 3.02585151e-04 2.01322463e-04
1.16432241e-04 1.15142744e-04 3.21896607e-04 1.80535658e-04
7.39190146e-05 5.10154335e-05 8.66236288e-06]

Two-related samples t-tets between TFR_open_alpha_C4 and TFR_closed_alpha_C4 pvalue
s: [1.56941441e-04 1.33620558e-05 1.20297974e-05 1.30838327e-05

2.23371919e-05 1.00187424e-05 5.70566546e-05 8.56392643e-05
1.35619574e-04 7.32070611e-05 1.80021848e-05 1.08722812e-04
1.74940902e-04 4.48259285e-05 8.76347296e-05 8.32795794e-05
8.71240302e-05 4.10262102e-04 9.60670469e-05 2.22080950e-04
5.34109326e-04 2.84215802e-04 1.30741420e-04 2.88203195e-04
3.38996479e-04 7.77962170e-04 1.65082897e-04 6.74216698e-05
1.54900143e-04 1.16578396e-04 3.50923197e-04 2.35751330e-05
2.42836881e-05 4.46780124e-05 2.50810013e-05 8.87767991e-05
6.61038931e-04 1.58919594e-04 2.21854609e-04 4.10710977e-04
2.21064596e-04 1.69219468e-04 1.59056024e-04 6.35487049e-05
6.79343817e-05 6.94412772e-05 3.05530128e-04 5.28135376e-04
3.98542800e-04 5.54894160e-04 3.40885848e-04 1.10603291e-04
1.09460037e-04 4.36296026e-04 1.94522481e-04 2.62594684e-04
4.84838874e-04 2.76570363e-04 4.02640974e-05 2.57684079e-06
3.14709886e-05 3.19233623e-04 1.29249124e-04 7.34743732e-05
7.17568355e-05 7.83368770e-05 4.66190678e-07 1.38535975e-06
4.17494153e-06 6.70997536e-07 2.08638835e-05 2.23048127e-04
6.96611877e-05 2.50610680e-05 4.08932803e-06 1.88736683e-04
3.86355948e-04 7.03941012e-05 2.01277104e-04 4.09724204e-04
5.93240250e-05 3.07853668e-05 6.33052908e-04 2.25086786e-03
1.53882018e-03 4.28655428e-04 1.50214271e-04 1.44652052e-04
1.17126472e-04 2.42708755e-04 1.18138760e-04 7.31363360e-04
1.31309333e-03 9.63612788e-04 2.62659495e-04 5.49808069e-05
3.58625320e-05 1.60470777e-04 9.91643763e-06 1.87174406e-05
1.32150158e-04 5.37225136e-04 8.89581116e-05 6.42034197e-05
1.13369416e-04 1.18382358e-04 2.94969437e-04 4.03901399e-04
1.16419073e-04 7.25023078e-05 1.09563657e-04 2.93872953e-04
4.00931516e-04 1.55673008e-04 4.50450043e-04 6.19310957e-04
3.49318499e-04 2.31328150e-04 5.82548575e-04 9.56361752e-04
1.41974565e-03 1.35677602e-03 8.23409888e-04 3.14336770e-04
4.84969346e-05 1.47264603e-04 2.99038611e-04 7.98265817e-05
3.91411546e-05 1.74024246e-05 1.50706611e-06 2.38421261e-05
1.86636402e-05 2.51653537e-05 1.39261463e-05 1.99838127e-06
3.10515409e-05 1.77399600e-04 9.69214547e-05 6.73730759e-05

8.00165976e-05 1.13701689e-04 2.50569342e-04 4.73346682e-05
2.10918727e-04 6.54365795e-04 3.08813222e-04 1.68336271e-04
4.72261339e-04 3.50843753e-04 5.61341948e-04]

Two-related samples t-tets between TFR_open_alpha_Pz and TFR_closed_alpha_Pz pvalue
s: [6.24508607e-02 2.35330440e-02 5.94380298e-03 3.42211231e-02

1.50203114e-02 1.65431909e-02 8.42543506e-04 6.54186990e-03
1.46272749e-02 2.81613422e-03 1.22620592e-02 1.72704457e-02
3.78583156e-02 1.54056098e-03 2.72439383e-04 1.24029323e-01
1.96629679e-02 5.43634934e-01 9.35493774e-01 1.15613801e-01
7.95601132e-02 2.07890032e-01 7.11033192e-03 8.59068777e-03
6.49513767e-02 3.95281895e-02 3.34128960e-02 5.86874248e-03
7.67626744e-04 1.42696571e-04 1.91949655e-04 1.56988612e-03
7.40747978e-04 2.20509084e-04 4.21027865e-03 4.42607343e-03
2.08717332e-03 1.55876463e-03 1.84090396e-03 3.63171425e-03
5.37732421e-03 3.42193856e-03 5.22149574e-03 5.22023349e-03
6.21774267e-04 8.35603191e-03 2.13594019e-02 1.02021632e-02
2.62054740e-02 4.43419048e-02 1.82982685e-02 1.78529807e-03
3.57349244e-03 9.24981497e-03 4.79893326e-02 7.57348554e-02
1.87782233e-02 9.70712105e-03 3.56190578e-02 2.43401394e-02
7.14448454e-03 1.50507714e-02 2.02273424e-02 4.96464327e-02
1.92008023e-02 7.49035235e-03 7.66742015e-02 8.81634108e-01
8.88450801e-01 3.59428168e-01 3.83787754e-01 7.98361201e-01
7.53290786e-01 5.11660696e-01 5.57097674e-01 3.48238793e-01
4.77199583e-02 1.37052493e-02 1.26914827e-02 3.83600741e-02
4.68664553e-02 2.37087789e-03 4.38082932e-04 1.58050770e-03
2.62795580e-03 1.05448157e-03 4.21017450e-03 7.96243471e-04
1.90128353e-03 1.03789806e-03 1.98408238e-03 1.65711383e-03
3.46463221e-03 7.39880602e-03 3.74368417e-03 2.71624637e-04
3.02190643e-05 1.46336044e-03 4.08398571e-03 7.99468775e-04
6.92408679e-05 5.56120637e-04 9.94004859e-05 1.67715896e-05
1.84936923e-03 2.51382312e-03 1.08365368e-03 6.84657875e-04
4.14391632e-03 2.78314999e-03 4.88918641e-03 3.49549765e-03
7.65485116e-03 1.46397684e-02 7.96513851e-03 3.40347744e-04
1.30204227e-02 1.64053346e-02 2.35579167e-02 1.54730718e-02
2.38482629e-02 5.66462483e-03 1.82512739e-02 9.88268247e-03
4.69121383e-03 6.37076028e-03 5.51119065e-02 7.31691753e-02
1.18089921e-02 1.75209694e-02 4.48501979e-02 1.44648873e-02
7.73168272e-04 8.50958954e-03 1.46370196e-03 7.31847000e-03
1.70422956e-02 5.32279183e-02 6.96764220e-03 7.91559971e-02
2.10419548e-01 4.66787674e-02 4.93705209e-02 1.06277334e-01
4.62241011e-03 3.85814706e-03 7.72498718e-04 4.55012374e-02
7.24716730e-02 3.69252935e-02 4.30652956e-02]

Two-related samples t-tets between TFR_open_alpha_P07 and TFR_closed_alpha_P07 pval
ues: [1.81506771e-05 7.68212801e-07 1.47288296e-07 1.06101057e-05

5.42934054e-06 8.74544615e-06 5.73202546e-06 7.38898153e-06
7.66119627e-06 3.94253361e-06 4.86949500e-07 4.44848107e-06
1.30692449e-05 8.96249724e-05 1.41400733e-04 7.33748324e-05
2.90111752e-04 5.32666269e-04 2.57104829e-04 2.77817175e-04
6.24681082e-04 5.41616705e-04 2.56240408e-04 2.01602736e-04
1.57775468e-04 3.42584087e-04 7.56444636e-04 3.63729620e-04
9.57859537e-05 1.30012008e-04 6.77186517e-04 3.53204202e-04

1.62739103e-04 6.04240858e-05 8.72393062e-05 2.29971095e-05
3.26585674e-05 2.53220854e-05 3.86674205e-05 5.74901782e-05
5.30841630e-05 1.13766148e-05 8.45049674e-06 2.44790555e-05
2.46178039e-05 2.36409711e-05 7.15606836e-05 7.08826640e-05
8.52709367e-06 2.77393603e-05 1.01183273e-04 2.44922594e-04
3.28783798e-04 1.19446455e-04 2.65229933e-04 3.71787870e-04
4.41892584e-04 1.90812141e-03 2.49062051e-03 2.15047955e-03
7.72774189e-04 8.13216521e-04 4.77588988e-04 9.49761018e-05
3.92589066e-04 8.42131024e-04 7.09378786e-04 2.04991939e-04
1.10609286e-04 9.71148502e-05 4.41998880e-05 4.33370327e-05
2.36291590e-05 2.08375479e-05 3.71591235e-04 1.12295509e-03
9.98548278e-04 2.43408566e-04 8.14902147e-05 9.47383582e-04
2.26314201e-03 1.25570208e-02 1.28809622e-02 3.53768464e-03
2.38179591e-03 1.89124960e-03 3.14675619e-03 3.45520073e-03
5.55354914e-03 6.85088551e-03 2.20356020e-03 3.03052716e-04
2.79370126e-04 2.99559199e-04 3.40499402e-04 2.29704928e-04
1.10438977e-04 5.79209664e-06 9.39091604e-06 1.19884853e-04
1.27430285e-04 1.16504829e-06 7.15520390e-06 2.30185461e-06
1.12738066e-05 9.48834248e-07 3.50886119e-05 4.51504341e-05
1.68735025e-06 4.25955544e-05 3.71209420e-05 6.13030698e-06
1.97637131e-07 2.19432634e-05 3.33888140e-05 1.69478068e-05
5.93840762e-07 1.32679301e-05 5.28284806e-05 6.20979400e-06
2.37397502e-06 3.70389641e-05 5.07342779e-05 4.32196645e-06
3.53825573e-05 2.34609774e-06 5.64128961e-07 2.68137876e-06
1.56262295e-05 9.64629397e-06 5.05756087e-05 3.04587707e-05
4.62172610e-05 1.03266841e-05 1.51090411e-06 1.49588009e-06
1.02133852e-06 3.30185601e-07 3.97853578e-05 9.07729528e-05
6.31396809e-05 1.25468830e-04 5.33189154e-04 3.06566317e-04
3.51871709e-04 6.59991106e-04 9.79635785e-04 3.37750614e-04
3.60886939e-04 2.91652597e-04 5.65772196e-04]

Two-related samples t-tets between TFR_open_alpha_Oz and TFR_closed_alpha_Oz pvalue

s: [1.31556000e-04 9.47159270e-05 1.43335537e-05 2.13220347e-06
7.99632419e-06 8.69338330e-06 7.24567409e-06 1.44296752e-05
3.62124896e-07 5.62723018e-06 2.35750323e-06 5.44648764e-07
3.12258263e-05 1.23049446e-04 5.24092350e-04 8.70453559e-04
1.91193596e-03 1.27059547e-03 6.97237515e-04 6.90620276e-04
1.41061577e-03 1.04726543e-03 9.42083489e-04 5.72802123e-04
1.52041581e-04 5.34593455e-06 1.99237823e-05 6.97863907e-05
5.54199995e-04 2.74503662e-04 2.27274335e-05 4.22205374e-06
8.40542802e-07 4.39222135e-06 2.45406336e-06 2.54992550e-05
2.52955173e-05 1.41464049e-05 5.70547681e-05 2.88744581e-04
4.09858227e-04 1.15481408e-04 1.12826243e-04 7.55962965e-05
3.27956993e-05 8.02961287e-06 5.60262014e-05 9.01184964e-05
3.60532725e-07 3.36547991e-04 6.87009816e-04 2.58134245e-04
3.96073468e-04 1.45040092e-04 1.10694541e-05 1.10234145e-04
4.69320609e-04 4.09698176e-04 6.60702191e-04 5.68071910e-04
1.20769890e-04 6.09129419e-05 6.67579577e-05 2.97974750e-05
1.90178480e-05 1.09888906e-05 7.89666238e-05 3.47852736e-04
4.28150102e-04 3.62573283e-04 5.70868775e-04 6.78278409e-04
2.38801509e-05 7.58511988e-05 1.07613630e-04 1.19395949e-04
4.05624494e-04 4.84437138e-04 5.39699775e-04 4.76643754e-04

4.37572198e-04 1.98344013e-04 1.97517451e-04 6.89794327e-05
8.05239917e-05 2.72598646e-05 1.70363341e-05 7.01492043e-06
1.43319712e-05 3.39206043e-04 6.58549793e-05 2.47172583e-05
4.68409260e-05 7.01986735e-06 3.12157060e-06 1.01157169e-05
8.10719288e-06 9.61530457e-05 2.45392375e-06 1.39290019e-05
1.07622605e-04 1.02702728e-05 1.29202642e-06 1.25528509e-05
6.12638291e-05 3.07924785e-05 1.21423573e-06 4.68478591e-05
1.08143264e-04 7.94563534e-06 1.02770617e-06 1.09378325e-05
6.46235499e-06 2.14411846e-05 3.39228230e-05 2.47041271e-05
6.18876193e-06 1.29683749e-05 8.20720056e-06 2.71670541e-05
1.46107406e-05 9.91630404e-05 4.73920303e-05 7.33397741e-05
2.31743454e-05 9.00160109e-06 9.09664921e-05 5.35882332e-05
7.07517973e-05 1.49277393e-04 1.69940196e-04 3.00017453e-05
3.18941491e-05 6.27963156e-05 2.96488914e-05 1.96870130e-05
3.87163757e-05 8.28269652e-07 2.69554202e-05 2.93567226e-04
3.40181663e-04 8.85304441e-04 7.37321306e-04 3.10142665e-04
6.31695685e-04 1.79711911e-03 5.73841683e-04 6.95801584e-04
4.76436868e-04 1.52422748e-04 3.81713519e-06]

Two-related samples t-tets between TFR_open_alpha_P08 and TFR_closed_alpha_P08 pval
ues: [1.20444150e-05 2.26091833e-05 2.46520684e-06 6.92812521e-07

2.83073193e-06 8.65840708e-06 1.03134775e-04 8.04790515e-06
6.65068475e-06 1.16553743e-05 1.06223987e-04 1.29691780e-05
1.89031991e-05 7.12634599e-05 5.32278232e-04 4.51460045e-04
6.82054438e-04 1.67105529e-03 2.56662990e-03 1.51079480e-03
1.28979523e-03 2.44370404e-03 6.21618358e-04 4.25559616e-05
4.11292912e-05 1.97555828e-04 5.92573612e-05 4.80826521e-05
1.84411067e-04 4.94143789e-04 2.79571633e-04 7.34507864e-05
1.34651460e-05 3.83766944e-05 8.32764932e-05 4.63901481e-05
1.71873117e-06 1.63229894e-05 3.40175183e-05 4.94708888e-05
4.50856768e-05 1.63605683e-04 3.62982735e-04 3.84191988e-05
1.22181483e-05 2.21440669e-06 3.88197057e-05 1.24115344e-05
4.24686669e-06 1.34003010e-04 2.03321208e-04 1.26751176e-05
1.70153918e-04 1.32004639e-04 1.17247987e-05 1.08439530e-05
1.14678076e-04 2.46492642e-04 1.30099031e-04 1.21262681e-04
4.39295958e-04 9.85448418e-05 6.57680879e-05 3.62500007e-05
5.37020738e-06 9.26666058e-06 1.44612150e-04 1.18034375e-04
9.92648535e-05 1.42805269e-04 9.00226372e-06 4.86221623e-05
2.69852123e-05 1.59285376e-05 1.77451907e-04 3.50196957e-04
9.78092357e-04 3.03158626e-04 4.43816435e-04 6.44560059e-04
2.18937009e-04 7.45751403e-05 4.55768568e-04 1.07836128e-03
8.49446925e-04 5.59062083e-04 4.78528795e-04 5.02560400e-04
1.61627869e-04 5.41885685e-06 6.09355696e-05 9.99659971e-05
1.33748228e-04 7.23017209e-05 5.53571661e-05 6.56848635e-06
1.49952989e-05 8.90265389e-05 4.41286366e-04 3.13373290e-04
1.94495487e-04 2.52406246e-04 1.56530380e-04 3.26038707e-04
1.75130665e-04 1.62711123e-04 1.36247660e-04 1.20208297e-04
2.36910233e-04 4.01978038e-04 3.52982275e-04 2.83681366e-04
5.66968692e-05 8.71384301e-05 1.86134655e-04 1.27819595e-05
1.07582489e-04 2.10525812e-04 1.86759913e-04 1.09693478e-04
8.14262108e-05 3.53757278e-05 1.20236485e-04 2.73080514e-04
1.38529823e-04 7.12478482e-05 7.98055385e-05 6.26108960e-04

```
1.50505877e-03 1.39734267e-03 8.51105010e-04 9.00940548e-05
2.36217032e-04 2.58506593e-04 1.69425186e-05 3.01095206e-05
9.04585369e-06 1.16803443e-05 7.84617125e-05 7.52077328e-04
5.18235813e-04 1.12728094e-03 3.69607854e-03 3.17889303e-03
1.94633133e-03 1.07912010e-03 9.40570608e-04 5.55236800e-05
2.22553237e-04 8.97987592e-04 6.18726946e-03]
```

In [51]:

```
for chan in channels:
    file_tfr_open = np.load("5. TFR/TFR_open_alpha_"+chan+".npy")
    file_tfr_closed = np.load("5. TFR/TFR_closed_alpha_"+chan+".npy")
    stat, pvalue = stats.ttest_rel(file_tfr_open, file_tfr_closed)
    print("Channel: ", chan)
    print("Max: ", max(pvalue))
    print("Mean: ", sum(pvalue)/len(pvalue))
    print()
    # print("Two-related samples t-tets between {} and {} pvalues:
    {}.format("TFR_open_alpha_"+chan, "TFR_closed_alpha_"+chan, pvalue))
```

Channel: Fz
Max: 0.002112145345528448
Mean: 0.00015410654362412498

Channel: C3
Max: 0.7017695803061158
Mean: 0.012353922868261227

Channel: Cz
Max: 0.0014631200104863597
Mean: 0.00010271331675488269

Channel: C4
Max: 0.0022508678574544106
Mean: 0.0002481056495185605

Channel: Pz
Max: 0.9354937739187069
Mean: 0.0656478343132574

Channel: P07
Max: 0.012880962209351947
Mean: 0.0005813822665416645

Channel: Oz
Max: 0.0019119359639428503
Mean: 0.00021199128209656613

Channel: P08
Max: 0.006187269464165089
Mean: 0.00037036955710529555

Most comparisons' p-values are statistically significant, and on average most channels show a significant difference at the .01 alpha level. The two exceptions are C3, still significant at the .05 alpha level (p-value=.0123) and Pz, not significant (p-value=.0656).

The End

Extra: Connectivity visualization between channels for Alpha band

! pip install utils

```
! pip install mne_connectivity
```

In [52]:

```
! pip install utils  
! pip install mne_connectivity
```

Collecting utils

Downloading utils-1.0.2.tar.gz (13 kB)

Preparing metadata (setup.py): started

Preparing metadata (setup.py): finished with status 'done'

Building wheels for collected packages: utils

Building wheel for utils (setup.py): started

Building wheel for utils (setup.py): finished with status 'done'

Created wheel for utils: filename=utils-1.0.2-py2.py3-none-any.whl size=13934 sha256=8cdd4392048e87e97c7a8b397898e4cded30770f528d2d9cb771e151441cc759

Stored in directory: c:\users\iamah\appdata\local\pip\cache\wheels\15\0c\b3\674ae8c5d91c642c817d4d630bd58faa316724b136844094d

Successfully built utils

Installing collected packages: utils

Successfully installed utils-1.0.2

Collecting mne_connectivity

Obtaining dependency information for mne_connectivity from https://files.pythonhosted.org/packages/5f/9c/b22094268040108fe3e6a6555831a61fe00a75d2c968958ca2afaa858ef7/mne_connectivity-0.6.0-py3-none-any.whl.metadata

Downloading mne_connectivity-0.6.0-py3-none-any.whl.metadata (10 kB)

Requirement already satisfied: mne>=1.6 in c:\users\iamah\anaconda3\lib\site-packages (from mne_connectivity) (1.6.1)

Collecting netCDF4>=1.6.5 (from mne_connectivity)

Obtaining dependency information for netCDF4>=1.6.5 from https://files.pythonhosted.org/packages/b6/66/6fcd815bf53b3e1653a02356e84ed23d191baaa844c5caad31d20a31cce8/netCDF4-1.6.5-cp311-cp311-win_amd64.whl.metadata

Downloading netCDF4-1.6.5-cp311-cp311-win_amd64.whl.metadata (1.8 kB)

Requirement already satisfied: numpy>=1.21 in c:\users\iamah\anaconda3\lib\site-packages (from mne_connectivity) (1.24.3)

Requirement already satisfied: pandas>=1.3.2 in c:\users\iamah\anaconda3\lib\site-packages (from mne_connectivity) (1.5.3)

Requirement already satisfied: scipy>=1.4.0 in c:\users\iamah\anaconda3\lib\site-packages (from mne_connectivity) (1.10.1)

Requirement already satisfied: tqdm in c:\users\iamah\anaconda3\lib\site-packages (from mne_connectivity) (4.65.0)

Collecting xarray>=2023.11.0 (from mne_connectivity)

Obtaining dependency information for xarray>=2023.11.0 from https://files.pythonhosted.org/packages/d4/b5/2af411546a4c2be8d6315bc01be055fbad367bc9c57dbd773ad05db62b1c/xarray-2024.1.1-py3-none-any.whl.metadata

Downloading xarray-2024.1.1-py3-none-any.whl.metadata (11 kB)

Requirement already satisfied: matplotlib>=3.5.0 in c:\users\iamah\anaconda3\lib\site-packages (from mne>=1.6->mne_connectivity) (3.7.1)

Requirement already satisfied: pooch>=1.5 in c:\users\iamah\anaconda3\lib\site-packages (from mne>=1.6->mne_connectivity) (1.8.0)

Requirement already satisfied: decorator in c:\users\iamah\anaconda3\lib\site-packages (from mne>=1.6->mne_connectivity) (5.1.1)

Requirement already satisfied: packaging in c:\users\iamah\anaconda3\lib\site-packages (from mne>=1.6->mne_connectivity) (23.0)

Requirement already satisfied: Jinja2 in c:\users\iamah\anaconda3\lib\site-packages (from mne>=1.6->mne_connectivity) (3.1.2)

Requirement already satisfied: lazy-loader>=0.3 in c:\users\iamah\anaconda3\lib\site-packages (from mne>=1.6->mne_connectivity) (0.3)

Collecting cftime (from netCDF4>=1.6.5->mne_connectivity)

Obtaining dependency information for cftime from https://files.pythonhosted.org/packages/a8/d2/75554106ea5bd1d44c73d354155395846d3833a5effe3a8660adde385288/cftime-1.6.3-cp311-cp311-win_amd64.whl.metadata

Downloading cftime-1.6.3-cp311-cp311-win_amd64.whl.metadata (8.8 kB)

Requirement already satisfied: certifi in c:\users\iamah\anaconda3\lib\site-packages (from netCDF4>=1.6.5->mne_connectivity) (2023.7.22)

Requirement already satisfied: python-dateutil>=2.8.1 in c:\users\iamah\anaconda3\lib\site-packages (from pandas>=1.3.2->mne_connectivity) (2.8.2)

Requirement already satisfied: pytz>=2020.1 in c:\users\iamah\anaconda3\lib\site-packages (from pandas>=1.3.2->mne_connectivity) (2022.7)

Requirement already satisfied: colorama in c:\users\iamah\anaconda3\lib\site-packages (from tqdm->mne_connectivity) (0.4.6)

Requirement already satisfied: contourpy>=1.0.1 in c:\users\iamah\anaconda3\lib\site-packages (from matplotlib>=3.5.0->mne>=1.6->mne_connectivity) (1.0.5)

Requirement already satisfied: cyclor>=0.10 in c:\users\iamah\anaconda3\lib\site-packages (from matplotlib>=3.5.0->mne>=1.6->mne_connectivity) (0.11.0)

Requirement already satisfied: fonttools>=4.22.0 in c:\users\iamah\anaconda3\lib\site-packages (from matplotlib>=3.5.0->mne>=1.6->mne_connectivity) (4.25.0)

Requirement already satisfied: kiwisolver>=1.0.1 in c:\users\iamah\anaconda3\lib\site-packages (from matplotlib>=3.5.0->mne>=1.6->mne_connectivity) (1.4.4)

Requirement already satisfied: pillow>=6.2.0 in c:\users\iamah\anaconda3\lib\site-packages (from matplotlib>=3.5.0->mne>=1.6->mne_connectivity) (9.4.0)

Requirement already satisfied: pyparsing>=2.3.1 in c:\users\iamah\anaconda3\lib\site-packages (from matplotlib>=3.5.0->mne>=1.6->mne_connectivity) (3.0.9)

Requirement already satisfied: platformdirs>=2.5.0 in c:\users\iamah\anaconda3\lib\site-packages (from pooch>=1.5->mne>=1.6->mne_connectivity) (2.5.2)

Requirement already satisfied: requests>=2.19.0 in c:\users\iamah\anaconda3\lib\site-packages (from pooch>=1.5->mne>=1.6->mne_connectivity) (2.31.0)

Requirement already satisfied: six>=1.5 in c:\users\iamah\anaconda3\lib\site-packages (from python-dateutil>=2.8.1->pandas>=1.3.2->mne_connectivity) (1.16.0)

Requirement already satisfied: MarkupSafe>=2.0 in c:\users\iamah\anaconda3\lib\site-packages (from jinja2->mne>=1.6->mne_connectivity) (2.1.1)

Requirement already satisfied: charset-normalizer<4,>=2 in c:\users\iamah\anaconda3\lib\site-packages (from requests>=2.19.0->pooch>=1.5->mne>=1.6->mne_connectivity) (2.0.4)

Requirement already satisfied: idna<4,>=2.5 in c:\users\iamah\anaconda3\lib\site-packages (from requests>=2.19.0->pooch>=1.5->mne>=1.6->mne_connectivity) (3.4)

Requirement already satisfied: urllib3<3,>=1.21.1 in c:\users\iamah\anaconda3\lib\site-packages (from requests>=2.19.0->pooch>=1.5->mne>=1.6->mne_connectivity) (1.26.16)

Downloading mne_connectivity-0.6.0-py3-none-any.whl (107 kB)

----- 0.0/107.2 kB ? eta -:-:-

----- 107.2/107.2 kB 6.1 MB/s eta 0:00:00

Downloading netCDF4-1.6.5-cp311-cp311-win_amd64.whl (6.6 MB)

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--- 0.6/6.6 MB 20.5 MB/s eta 0:00:01

----- 1.8/6.6 MB 23.1 MB/s eta 0:00:01

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Downloading xarray-2024.1.1-py3-none-any.whl (1.1 MB)
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----- 1.1/1.1 MB 34.5 MB/s eta 0:00:00
Downloading cftime-1.6.3-cp311-cp311-win_amd64.whl (188 kB)
----- 0.0/188.2 kB ? eta -:--:--
----- 188.2/188.2 kB ? eta 0:00:00
Installing collected packages: cftime, netCDF4, xarray, mne_connectivity
  Attempting uninstall: xarray
    Found existing installation: xarray 2023.6.0
    Uninstalling xarray-2023.6.0:
      Successfully uninstalled xarray-2023.6.0
Successfully installed cftime-1.6.3 mne_connectivity-0.6.0 netCDF4-1.6.5 xarray-2024.1.1

```

```

In [53]: import utils
import mne_connectivity
from mne_connectivity import spectral_connectivity_epochs
from mne_connectivity.viz import plot_connectivity_circle,
plot_sensors_connectivity
from mne.viz import circular_layout
from scipy import stats

```

```

In [54]: # Eyes open:

```

```

In [55]: epoch_E0_one = mne.read_epochs("4. Epoch/Open/Open_Eyes_1-
open_eyes_epo.fif", preload= True)

```

```

Reading C:\Users\IamAh\Desktop\UNIVERSITY - CSAI\Cognitive Neuroscience\M001_Cognitive_Neuroscience_Practical_Material\EEG_practical\4. Epoch\Open\Open_Eyes_1-open_eyes_epo.fif ...
Found the data of interest:
  t =  -1000.00 ...   5000.00 ms
  0 CTF compensation matrices available
Not setting metadata
11 matching events found
No baseline correction applied
0 projection items activated

```

```

In [56]: e_o = []

```

```
In [57]: con_E0 = spectral_connectivity_epochs(epoch_E0_one, names=channels,
method='dpli', indices=None, sfreq=250, mode='multitaper',
          fmin=8, fmax=12, fskip=0, faverage=True,
          tmin=None, tmax=None, mt_bandwidth=None, mt_adaptive=False,
          mt_low_bias=True, cwt_freqs=None,
          cwt_n_cycles=7, block_size=1000, n_jobs=1, verbose=None)
e_o.append(con_E0)
```

Adding metadata with 3 columns

Connectivity computation...

only using indices for lower-triangular matrix

computing connectivity for 28 connections

using t=-1.000s..5.000s for estimation (1501 points)

frequencies: 8.2Hz..12.0Hz (24 points)

connectivity scores will be averaged for each band

Using multitaper spectrum estimation with 7 DPSS windows

the following metrics will be computed: DPLI

computing cross-spectral density for epoch 1

computing cross-spectral density for epoch 2

computing cross-spectral density for epoch 3

computing cross-spectral density for epoch 4

computing cross-spectral density for epoch 5

computing cross-spectral density for epoch 6

computing cross-spectral density for epoch 7

computing cross-spectral density for epoch 8

computing cross-spectral density for epoch 9

computing cross-spectral density for epoch 10

computing cross-spectral density for epoch 11

assembling connectivity matrix

[Connectivity computation done]

```
In [58]: epoch_E0_two = mne.read_epochs("4. Epoch/Open/Open_Eyes_2-
open_eyes_epo.fif", preload=True)
```

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Found the data of interest:

t = -1000.00 ... 5000.00 ms

0 CTF compensation matrices available

Not setting metadata

11 matching events found

No baseline correction applied

0 projection items activated

```
In [59]: con_E0 = spectral_connectivity_epochs(epoch_E0_two, names=channels,
method='dpli', indices=None, sfreq=250, mode='multitaper',
fmin=8, fmax=12, fskip=0, faverage=True,
tmin=None, tmax=None, mt_bandwidth=None, mt_adaptive=False,
mt_low_bias=True, cwt_freqs=None,
cwt_n_cycles=7, block_size=1000, n_jobs=1, verbose=None)
e_o.append(con_E0)
```

Adding metadata with 3 columns

Connectivity computation...

only using indices for lower-triangular matrix

computing connectivity for 28 connections

using t=-1.000s..5.000s for estimation (1501 points)

frequencies: 8.2Hz..12.0Hz (24 points)

connectivity scores will be averaged for each band

Using multitaper spectrum estimation with 7 DPSS windows

the following metrics will be computed: DPLI

computing cross-spectral density for epoch 1

computing cross-spectral density for epoch 2

computing cross-spectral density for epoch 3

computing cross-spectral density for epoch 4

computing cross-spectral density for epoch 5

computing cross-spectral density for epoch 6

computing cross-spectral density for epoch 7

computing cross-spectral density for epoch 8

computing cross-spectral density for epoch 9

computing cross-spectral density for epoch 10

computing cross-spectral density for epoch 11

assembling connectivity matrix

[Connectivity computation done]

```
In [60]: e_o = e_o[0].append(e_o[1])
```

```
In [61]: print(e_o)
```

```
<SpectralConnectivity | freq : [10.076616, 10.076616], , nave : 11, nodes, n_estima
ted : 8, 64, ~50 kB>
```

```
In [62]: conmat_E0 = con_E0.get_data(output='dense')[:, :, 0]
```

```
In [63]: # Eyes closed:
```

```
In [64]: epoch_EC_one = mne.read_epochs("4. Epoch/Closed/Closed_Eyes_1-
closed_eyes_epo.fif", preload= True)
```

```
Reading C:\Users\IamAh\Desktop\UNIVERSITY - CSAI\Cognitive Neuroscience\M001_Cognitive_Neuroscience_Practical_Material\EEG_practical\4. Epoch\Closed\Closed_Eyes_1-closed_eyes_epo.fif ...
```

```
Found the data of interest:
```

```
t = -1000.00 ... 5000.00 ms
```

```
0 CTF compensation matrices available
```

```
Not setting metadata
```

```
11 matching events found
```

```
No baseline correction applied
```

```
0 projection items activated
```

```
In [65]: e_c = []
```

```
In [66]: con_EC = spectral_connectivity_epochs(epoch_EC_one, names=channels,
method='dpli', indices=None, sfreq=250, mode='multitaper',
          fmin=8, fmax=12, fskip=0, faverage=True,
          tmin=None, tmax=None, mt_bandwidth=None, mt_adaptive=False,
          mt_low_bias=True, cwt_freqs=None,
          cwt_n_cycles=7, block_size=1000, n_jobs=1, verbose=None)
e_c.append(con_EC)
```

```
Adding metadata with 3 columns
```

```
Connectivity computation...
```

```
only using indices for lower-triangular matrix
```

```
computing connectivity for 28 connections
```

```
using t=-1.000s..5.000s for estimation (1501 points)
```

```
frequencies: 8.2Hz..12.0Hz (24 points)
```

```
connectivity scores will be averaged for each band
```

```
Using multitaper spectrum estimation with 7 DPSS windows
```

```
the following metrics will be computed: DPLI
```

```
computing cross-spectral density for epoch 1
```

```
computing cross-spectral density for epoch 2
```

```
computing cross-spectral density for epoch 3
```

```
computing cross-spectral density for epoch 4
```

```
computing cross-spectral density for epoch 5
```

```
computing cross-spectral density for epoch 6
```

```
computing cross-spectral density for epoch 7
```

```
computing cross-spectral density for epoch 8
```

```
computing cross-spectral density for epoch 9
```

```
computing cross-spectral density for epoch 10
```

```
computing cross-spectral density for epoch 11
```

```
assembling connectivity matrix
```

```
[Connectivity computation done]
```

```
In [67]: epoch_EC_two = mne.read_epochs("4. Epoch/Closed/Closed_Eyes_2-closed_eyes_epo.fif", preload=True)
```

```
Reading C:\Users\IamAh\Desktop\UNIVERSITY - CSAI\Cognitive Neuroscience\M001_Cognitive_Neuroscience_Practical_Material\EEG_practical\4. Epoch\Closed\Closed_Eyes_2-closed_eyes_epo.fif ...
```

```
Found the data of interest:
```

```
t = -1000.00 ... 5000.00 ms
```

```
0 CTF compensation matrices available
```

```
Not setting metadata
```

```
11 matching events found
```

```
No baseline correction applied
```

```
0 projection items activated
```

```
In [68]: con_EC = spectral_connectivity_epochs(epoch_EC_two, names=channels,
method='dpli', indices=None, sfreq=250, mode='multitaper',
fmin=8, fmax=12, fskip=0, faverage=True,
tmin=None, tmax=None, mt_bandwidth=None, mt_adaptive=False,
mt_low_bias=True, cwt_freqs=None,
cwt_n_cycles=7, block_size=1000, n_jobs=1, verbose=None)
e_c.append(con_EC)
```

```
Adding metadata with 3 columns
```

```
Connectivity computation...
```

```
only using indices for lower-triangular matrix
```

```
computing connectivity for 28 connections
```

```
using t=-1.000s..5.000s for estimation (1501 points)
```

```
frequencies: 8.2Hz..12.0Hz (24 points)
```

```
connectivity scores will be averaged for each band
```

```
Using multitaper spectrum estimation with 7 DPSS windows
```

```
the following metrics will be computed: DPLI
```

```
computing cross-spectral density for epoch 1
```

```
computing cross-spectral density for epoch 2
```

```
computing cross-spectral density for epoch 3
```

```
computing cross-spectral density for epoch 4
```

```
computing cross-spectral density for epoch 5
```

```
computing cross-spectral density for epoch 6
```

```
computing cross-spectral density for epoch 7
```

```
computing cross-spectral density for epoch 8
```

```
computing cross-spectral density for epoch 9
```

```
computing cross-spectral density for epoch 10
```

```
computing cross-spectral density for epoch 11
```

```
assembling connectivity matrix
```

```
[Connectivity computation done]
```

```
In [69]: e_c = e_c[0].append(e_c[1])
```

```
In [70]: print(e_c)
```

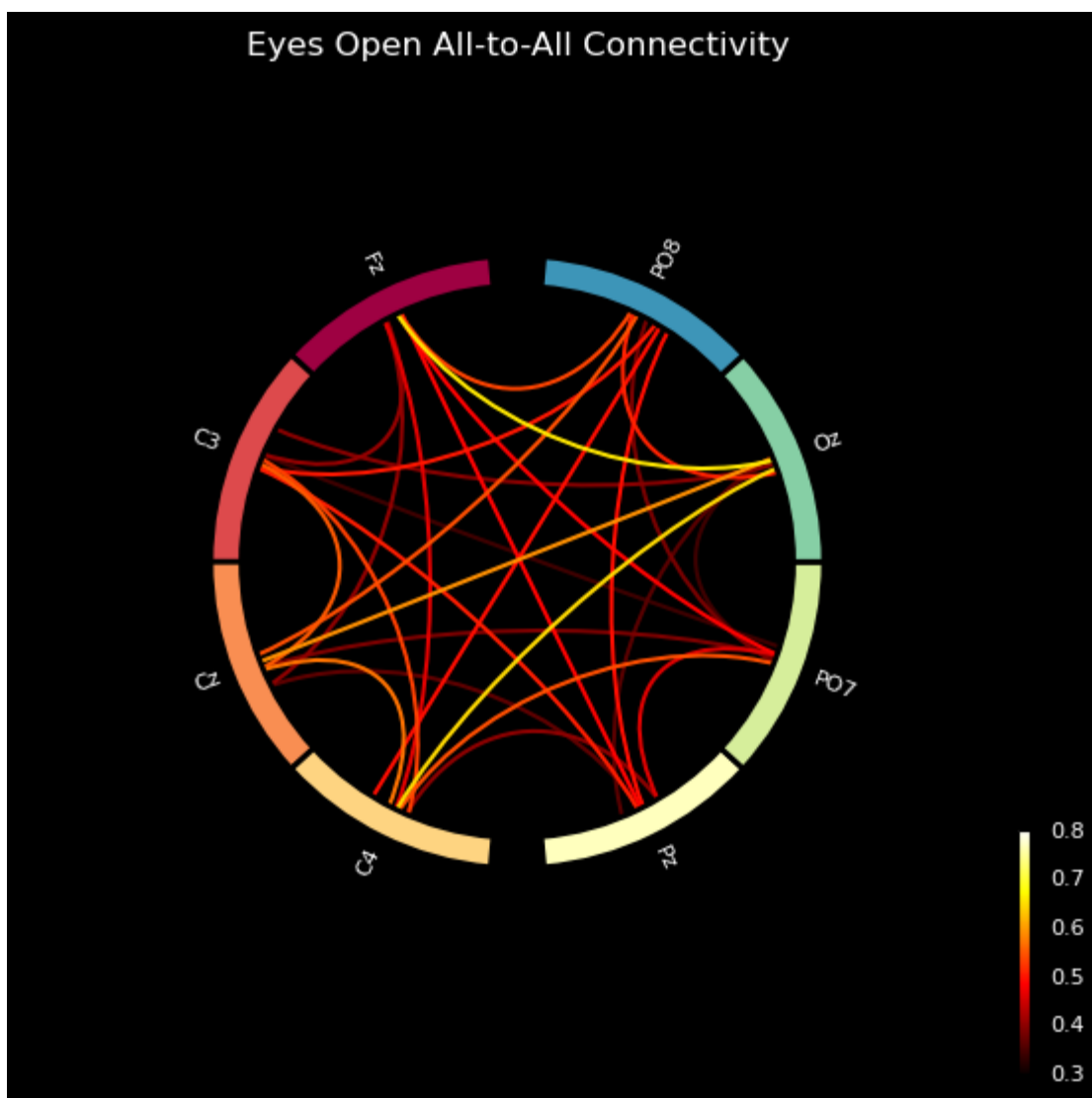
```
<SpectralConnectivity | freq : [10.076616, 10.076616], , nave : 11, nodes, n_estimated : 8, 64, ~50 kB>
```

```
In [71]: conmat_EC = con_EC.get_data(output='dense')[:, :, 0]
```

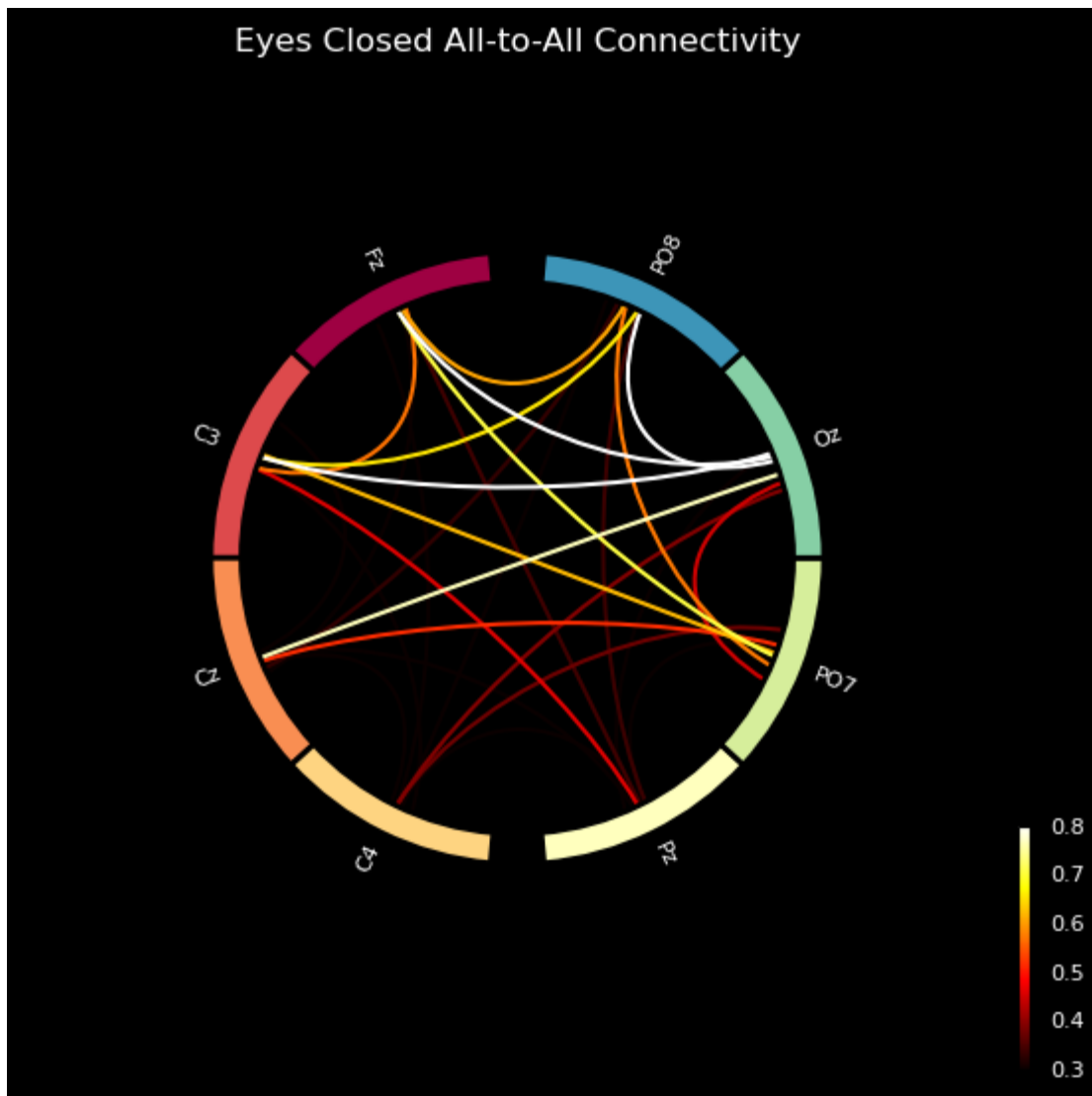
```
In [72]: # Connectivity plot
```

```
In [73]: node_angles = circular_layout(channels, channels, start_pos=90,  
                                       group_boundaries=[0, len(channels) // 2])
```

```
In [74]: fig, ax = plt.subplots(figsize=(8, 8), facecolor='black',  
                                subplot_kw=dict(polar=True))  
plot_connectivity_circle(conmat_EC, node_names = channels,  
                        node_angles=node_angles, node_colors=None,  
                        title='Eyes Open All-to-All Connectivity',  
ax=ax, vmin = 0.3, vmax = 0.8)  
fig.tight_layout()
```



```
In [75]: fig, ax = plt.subplots(figsize=(8, 8), facecolor='black',
                        subplot_kw=dict(polar=True))
plot_connectivity_circle(conmat_EC, node_names = channels,
                        node_angles=node_angles, node_colors=None,
                        title='Eyes Closed All-to-All Connectivity',
                        ax=ax, vmin = 0.3, vmax = 0.8)
fig.tight_layout()
```

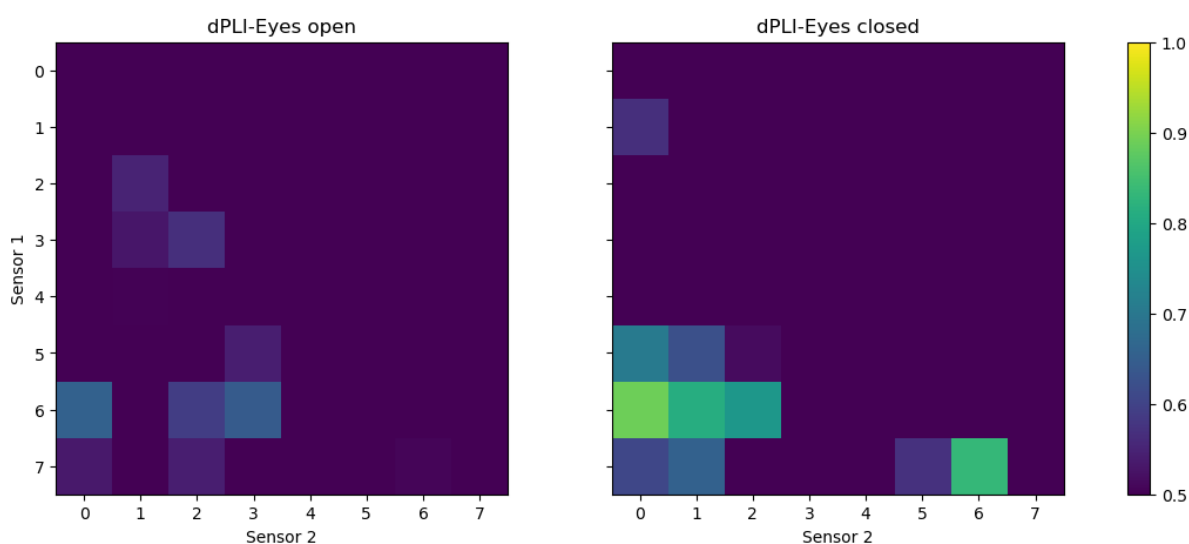


Phase lag index

```
In [76]: fig, axs = plt.subplots(1, 2, figsize=(14, 5), sharey=True)
axs[0].imshow(con_E0.get_data('dense'), vmin=0.5, vmax=1)
axs[0].set_title("dPLI-Eyes open")
axs[0].set_ylabel("Sensor 1")
axs[0].set_xlabel("Sensor 2")

im = axs[1].imshow(con_EC.get_data('dense'), vmin=0.5, vmax=1)
axs[1].set_title("dPLI-Eyes closed")
axs[1].set_xlabel("Sensor 2")

fig.colorbar(im, ax=axs.ravel())
plt.show()
```



In []: